# Silent Evidence Defects in the Wild: A Registered Replay of a Real Purchase-to-Pay Process

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## Abstract

Of the 155,534 analysed invoice clearings of a real purchase-to-pay event log (Section 4 gives the registered exclusions), 17,748 (11.41%, clustered bootstrap 95% interval [10.94%, 11.92%]) were made on evidence that failed at least one of seven registered predicates, carrying 19.41% of the cleared value ([16.04%, 23.09%]). Of the clearings with a failed predicate, 16,478 fail D4 with the line's creation as the last approval-bearing event, no explicit approval or release event being recorded on the line before the clearing; counting only changes after an explicit approval, an exploratory reading, exposure is 0.90% of clearings [0.67%, 1.19%] and 5.65% of value [3.67%, 8.13%]. We replay every analysed clearing of the BPI Challenge 2019 log through a deterministic min() gate declared before the data were read, over the evidence reconstructed from the log at clearing time. Clearings made on evidence that failed on the goods receipt (absent or late, D1/D2) were followed by a corrective event within 60 days at a rate of 5.64%, against 0.93% for three-way clearings with every applicable predicate passing (difference 0.0470, Holm-adjusted $p = 4.53 \times 10^{-4}$); clearings made after a change of price or quantity since approval (D4 fail), at 1.91% against 0.96% for D4-pass clearings (difference 0.0094, Holm-adjusted $p = 3.48 \times 10^{-7}$). On a registered sample of lines with a failed predicate and of clean lines, two model families shown the ERP view held 100.0% and 99.6% of the first and 53.0% and 57.3% of the second; on the sample lines the log lets us classify, people held 33.0% and 34.0%, where a hold is a recorded payment block or, in nearly every case, an inferred non-release within 60 days, not a recorded decision; the gate's verdict holds the first and allows the second by construction. A line's status at the decision point follows a reading of D5 (decision D-C-2 B) that the author fixed after E3 and before any model call (Section 4). Paper 5's compiler, given the three declared losses of the registration, does not reproduce the registered contract in any item category. The coverage matrix states which defect classes this log cannot expose; the authority defect class (D5) has zero incidence here. No causal claim is made.

## 1 Introduction

Earlier papers of this programme showed the mechanism of a silent evidence defect, a decision taken on evidence that fails a metadata-borne predicate, on constructed domains, and made no prevalence claim. The six papers are on arXiv: paper 1, SARC, on whether the agent may act (Besanson 2026a); paper 2, Green SARC, on its budget (Besanson 2026b); paper 3, SARC-DQ, on the evidence it admits (Besanson 2026c); paper 4 on composing several gates on one action (Besanson 2026d); paper 5 on deriving the contract, the compiler (Besanson 2026e); and paper 6 on facts read from documents (Besanson 2026f). This paper is the number, on one real process.

The unit is an invoice clearing in the BPI Challenge 2019 purchase-to-pay log. For each of the 155,534 analysed clearings (the exclusion table of Section 4) we reconstruct the evidence recorded on the purchase-order line at clearing time, evaluate the registered predicates D1 to D7, record what the gate would have held, and read what followed in the log. We report exposure in count and in anonymised EUR, the corrective events that follow a failed predicate, the reach of the gate, and, on a registered sample, what two model families decide on the same ERP view beside the people in the log and the gate. Everything was registered before the log was parsed beyond its published counts (tag `prereg-p7-v1`) and amended once after the pilot (v1.1, tag

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`prereg-p7-v1.1`), with five deviations under v1.1: two decided after E2 and E3 and before E4, and three found in review (corrected descriptions in the coverage matrix, the E7 bootstrap replicates with an empty group discarded, and the request format of the Jev family in E4); all are listed in Section 4.

What this paper claims is in the claim table of Appendix B. What it does not claim is stated in the repository's claims file and repeated here verbatim:

- No causal claim. Correctives are a proxy, observational, confounded by vendor, category and practice.
- No loss figure. A corrected payment is not money gone; recovery is not in the log.
- No claim that the gate stops anything. Held lines go somewhere the log does not show.
- No claim about the company or any vendor. No company, subsidiary or vendor is named in any finding.
- No agent acting on the live system and no agent with tools.
- No readiness threshold.

"No claim about the company" means that no company, subsidiary or vendor is named or characterised; the measured process is the subject of every finding.

## 2 Related work

This paper does not introduce conformance checking, the three-way match or rework analysis. It replays each clearing as a gate decision over the evidence reconstructed from the event log at clearing time, measures how often that evidence was defective and what followed, and puts people, models and the gate on the same lines. The defect classes are the predicates of Section 3.

**Conformance and compliance checking.** Checking recorded behaviour against a normative model or rule set is established (van der Aalst 2016; Ly et al. 2015), and declarative constraints in the Declare family take a process model as the conjunction of its constraints (Pesic et al. 2007). The predicates D1 to D7 are compliance rules in this sense, and taking the decision as their conjunction is Declare's. Synthesising a sufficient contract from declared losses is paper 5 of this programme (Besanson 2026e); the compiler used in Section 5.5 is paper 5's. Detecting that one resource performed two activities that must be separated is a standard compliance monitoring functionality (Ly et al. 2015); predicate D5 is that check.

**The BPI Challenge 2019 reports.** The challenge posed the process owner's compliance questions on this very log, including clearing before goods receipt and the three-way match per item category (ICPM 2019). The submitted reports measured basic violation rates and are prior art for every basic rate reported here; among them Diba et al. 2019 quantify receipt-before-clearing violations with their value, and Porter et al. 2019 report compliance per item category both by number and by value. Measuring compliance by count and by value is therefore not claimed; what is claimed is the tabulation per clearing and per defect class, with clustered intervals. The three-way match itself is a textbook application control (PCAOB 2004); Section 5.5 asks whether paper 5's compiler re-derives it from declared losses.

**Deviation and outcome.** Relating a deviation to a later outcome in the same case is established: the extension of process mining to outcomes associated with deviations (Peleg et al. 2008), outcome-oriented predictive process monitoring (Teinemaa et al. 2019) and deviance mining (Nguyen et al. 2016). On this log, Porter et al. 2019 observe cancellations of invoice receipts long after clearing as a case-study observation. Corrective events as rework signals are established on this log too: Meyer zu Wickern et al. 2019 count change, cancel and delete activities as rework, and Badakhshan et al. 2019 list vendor debit memos among rework activities. The outcome proxy of Section 3 is a rework signal of this kind; H1 to H3 are registered conditional-consequence studies of the deviation-to-outcome shape.

**Detectability.** That some deviations leave no trace in an event log is established (Outmazgin and Soffer 2013). The registered per-class coverage matrix of Section 6 is claimed; the principle is not.

**Deciders on the same cases.** Putting different deciders on the same cases is established: a human reviewer, a model agent and learned rules on the same accounts (Palma et al. 2026), language models against audit baselines on the same ledgers (Kadir et al. 2025), and algorithms against bank managers (Harding and Vasconcelos 2022). None of these uses a deterministic contract declared before the data were read, or a human arm read from a process event log; Section 5.4 does both, with one qualification: the status of a line at the

decision point follows a reading of D5 (decision D-C-2 B) that the author fixed after E3 and before any model call (Section 4).

**The BPI Challenge 2020 reports.** The 2020 challenge posed questions on the approval chain of travel declarations (ICPM 2020b); two reports received awards (ICPM 2020a; Pufahl et al. 2020; Kas et al. 2020). They are prior art for the basic rates of the replication in Section 5.7.

## 3 Formal core

**Lines, clearings and evidence.** A line is one case of the log: one purchase document and item. A clearing is one `Clear Invoice` event on a line at time t_c. The evidence at t_c is every event on the line with timestamp at most t_c, excluding the clearing itself, as the registration defines "before t_c" (section 2 of `prereg-p7-v1`, unchanged in v1.1); events sharing a timestamp count as available, and counts rather than event order decide receipts, blocks and deletions, so the verdict does not depend on the order of events with equal timestamps (a property test of the replay engine).

**Category and contract.** The case flags select the contract, as the data record states the rules:

| category | GR-based invoice verification | goods receipt | contract |
|---|---|---|---|
| three-way, invoice after receipt (`3-way`) | true | true | D1 to D7 |
| three-way, invoice before receipt (`3-way`) | false | true | D1 to D7 |
| two-way (`2-way`) | false | false | D3 to D7 |
| consignment | any | any | excluded |

**Predicates.** Each predicate takes the value pass, fail or unknown; unknown means a required input is missing. Each predicate checks one SARC-DQ dimension (the table). The contract as a whole is checked with paper 5's own sufficiency function over this log's observed states (E5, `results/e5/sufficiency.json`, Section 5.5): it is sufficient for the three declared losses in the two-way category, and in the two three-way categories for L1 and L3 but not for L2; it is minimal in no category.

| id | name | dimension | fails when, on the line before t_c |
|---|---|---|---|
| D1 | receipt present | presence | no uncancelled goods receipt (receipts minus receipt cancellations at most zero) |
| D2 | receipt before clearing | freshness, order | no uncancelled receipt precedes t_c |
| D3 | invoice present | presence | no invoice receipt |
| D4 | unchanged since approval | validity | a price or quantity change after the last approval-bearing event |
| D5 | distinct actors | authority | the actor of the clearing equals an actor of the earliest creation (unknown if either is missing) |
| D6 | block respected | scope | payment blocks set minus blocks removed greater than zero |
| D7 | not deleted | validity | deletions plus blocks minus reactivations of the line greater than zero |

Under the registered reading of D4, the line's earliest creation (`Create Purchase Order Item`) is an approval-bearing event, so D4 flags any later price or quantity change on a line with no later approval or release event. The paper reports the registered reading as primary, and a strict reading, which anchors D4 on explicit approval or release events only, as exploratory (Section 5.1). A failed registered predicate is not a

confirmed violation of the company's approval policy: whether a change after creation required re-approval is not recorded in the log.

At t_c, D1 and D2 have the same truth value; they are kept apart because they carry two dimensions. For reporting only, a D1/D2 failure is assigned to a defect class by what the log records later: the stale receipt (D2 class) if an uncancelled receipt is recorded after t_c and by the observation end, the absent receipt (D1 class) otherwise.

**Verdict.** The verdict is min() over the category's contract with pass as one and fail or unknown as zero: the gate allows a clearing if and only if every applicable predicate passes, and otherwise would have held it. A clearing is defective if at least one applicable predicate fails; one whose only non-pass value is unknown is held but not defective, and is counted separately. "Defect" is used only for a failed predicate.

**Human hold.** A line is held by people if a payment block is recorded before its first clearing, or if no clearing is recorded within W days of its first invoice receipt; a clearing before the first invoice receipt counts as a release.

**D5 at a decision point that is not a clearing.** D5 compares the actor of the clearing with an actor of the line's earliest creation. At a decision point with no clearing there is no clearing actor: the predicate has no argument and is not applicable (decision D-C-2 B), which is not the "unknown" of an empty actor field on a recorded clearing. Invariant to this reading: E1, E2, every registered test and the Holm table, all computed at actual clearings. Dependent on it: the E3 human-hold slot and the E4 sample, which classify lines at t*. Both readings are reported (Section 5.3).

**Outcome proxy and lag bands.** O = 1 for a clearing if a registered corrective activity (an invoice receipt or goods receipt cancelled, a price or quantity changed, the line deleted, a vendor debit memo, a subsequent invoice) is recorded on the line within W after t_c; W = 60 days, with W = 30 days as a descriptive sensitivity. For a D1/D2 failure, the lag from t_c to the first uncancelled receipt falls in one of the registered bands [0, 1], (1, 7], (7, 30] days, over 30 days, or never.

**Evidence set.** One record per analysed clearing carries the line id (hashed in every export), category, t_c, every predicate input and value, the verdict, the defect classes, O at both windows, the cleared and line values, and the SHA-256 of the canonical serialisation of the line's events up to t_c.

## 4 Data and registration

**Data.** The primary log is the BPI Challenge 2019 record; the replication uses the two declaration logs of the BPI Challenge 2020 collection. All three are by Boudewijn van Dongen on 4TU.ResearchData; each file was checked against the hash the registration records.

| record | DOI | licence | file | cases | events | SHA-256 |
|---|---|---|---|---|---|---|
| BPI Challenge 2019 | doi:10.4121/uuid:d06aff4b-79f0-45e6-8ec8-e19730c248f1 | CC BY 4.0 | `BPI_Challenge_2019.xes` | 251,734 | 1,595,923 | `af63bc687fc4152f2123b05c3af7772b37ef3fce2d3f67f812666c9e356baae7` |
| BPI Challenge 2020: International Declarations | doi:10.4121/uuid:2bbf8f6a-fc50-48eb-aa9e-c4ea5ef7e8c5 | CC BY-NC 4.0 | `International Declarations.xes.gz` | 6,449 | 72,151 | `b527c8d06255f4312c6ce8f36f7f4110243d84fd6ca48583cd9b80896c601494` |
| BPI Challenge 2020: Domestic Declarations | doi:10.4121/uuid:3f422315-ed9d-4882-891f-e180b5b4feb5 | CC BY-NC 4.0 | `DomesticDeclarations.xes.gz` | 10,500 | 56,437 | `0bc3312acf0b9ee5ae6f4ebe9df204005ef11665d62f2d21dd04995161d7532b` |

Values in the 2019 log are anonymised by a linear translation that respects zero; we report them as anonymised EUR.

**Window, observation end and frame.** A line is in the window if its earliest `Create Purchase Order Item` event lies in calendar 2018. The observation end T_end is the latest timestamp in the last calendar month that carries at least 1,000 events; here T_end is 2019-01-18 and 9 later events are excluded. A line is framed if it survives the exclusions below and its first clearing, or its creation if it has none, lies at least W before T_end; the analysed clearings are every clearing on a framed line at least W before T_end, so every outcome window is complete. The lines drawn for the pilot (E0) are removed before any estimate. As registered, the `Item Category` string disagrees with the flag-derived category on 0 of 251,734 lines (section 3.1; `results/e1/category-crosscheck.json`); every analysed clearing took its value from the clearing event itself (155,534 clearings; 0 fell back to the invoice receipt and 0 to the line value, section 3.4; `results/e1/value-fallback.json`); and the seeded 1,000-record sample of the evidence set, identifiers hashed, is exported as `results/e1/evidence-sample.jsonl` (section 3.9).

| step | lines |
|---|---|
| lines in the log | 251,734 |
| 2.1 (1) no Create Purchase Order Item | 0 |
| 2.1 (2) consignment | 14,498 |
| 3.1 category undefined by the flags | 0 |
| 2.1 (3) zero or missing line value | 1,887 |
| 2.1 (4) more than K = 12 clearings | 78 |
| 2.1 (5) outside the window | 376 |
| 2.1 (5) failing the frame condition | 53,731 |
| framed lines | 181,164 |
| E0 pilot lines removed | 500 |
| framed lines analysed | 180,664 |
| analysed clearings | 155,534 |
| events after T_end excluded | 9 |

**Registration.** The registration was tagged `prereg-p7-v1` before any parse of the log beyond the published counts. The pilot fired registered stop conditions (the latest timestamp lay outside the registered range, and a mapped attribute is named differently in the log than on the record page); the registration was amended and tagged `prereg-p7-v1.1`, and the pilot was drawn again, before any estimate. Every deviation, with the decision that records its reason:

- p7-v1 to p7-v1.1: T_end rule.
- p7-v1 to p7-v1.1: Mapping attribute name.
- p7-v1 to p7-v1.1: Section 3.6, explicit sentence.
- Under p7-v1.1: H3 convention.
- Under p7-v1.1: D5 applicability at a non-clearing decision point.
- Under p7-v1.1: Description corrections in COVERAGE.md, no predicate changes.
- Under p7-v1.1: E7 bootstrap replicates with an empty group discarded.
- Under p7-v1.1: E4 Jev request format.
- Under p7-v1.1: Exploratory timeline arm: the clearing at t* was shown.

The H3 convention and the D5 reading were the author's decisions after E2 and E3 and before any model call (decision D-C-2 in the decision log). The response logs date the calls: D-C-2 was committed at 2026-10-25 18:21:56 UTC (`7aea13f`) and the E4 sample drawn at 2026-10-25 18:39:34 UTC (`5ffd887`); the first request was sent at 2026-10-25 18:42:18 UTC for Jev and at 2026-10-25 18:46:32 UTC for Haiku (`results/e4/timeline.json`; request times as the client wrote them before sending, commit times as git records them). The prior-art fence was reviewed externally after the tag `prereg-p7-v1` and before `prereg-p7-v1.1`, after two machine sweeps whose output is committed unedited.

## 5 Results

### 5.1 Exposure (E1, registered)

| defect class | clearings | share | Wilson 95% | bootstrap 95% | share of value | bootstrap 95% |
|---|---|---|---|---|---|---|
| D1 class (absent receipt) | 610 | 0.39% | [0.36%, 0.42%] | [0.30%, 0.50%] | 0.12% | [0.08%, 0.17%] |
| D2 class (stale receipt) | 82 | 0.05% | [0.04%, 0.07%] | [0.02%, 0.10%] | 0.12% | [0.03%, 0.27%] |
| D3 | 362 | 0.23% | [0.21%, 0.26%] | [0.15%, 0.33%] | 0.03% | [0.02%, 0.05%] |
| D4 | 17,157 | 11.03% | [10.88%, 11.19%] | [10.55%, 11.53%] | 19.24% | [15.88%, 22.89%] |
| D5 | 0 | 0.00% | [0.00%, 0.00%] | [0.00%, 0.00%] | 0.00% | [0.00%, 0.00%] |
| D6 | 4 | 0.00% | [0.00%, 0.01%] | [0.00%, 0.01%] | 0.03% | [0.00%, 0.10%] |
| D7 | 19 | 0.01% | [0.01%, 0.02%] | [0.01%, 0.02%] | 0.01% | [0.00%, 0.02%] |
| at least one defect | 17,748 | 11.41% | [11.25%, 11.57%] | [10.94%, 11.92%] | 19.41% | [16.04%, 23.09%] |
| held on an unknown only (not a defect) | 0 | 0.00% | [0.00%, 0.00%] | [0.00%, 0.00%] | 0.00% | [0.00%, 0.00%] |

Figure 1 shows the shares with their intervals. Shares of clearings carry a Wilson and a purchase-document-clustered bootstrap 95% interval (B = 10,000, seed 20261201); shares of value carry the clustered interval. The denominator is every analysed clearing, and a share counts the clearings made on evidence that failed the predicate; D1 and D2 classes apply to 155,346 clearings in the three-way categories. A change after approval (D4) is the largest defect class, 17,157 clearings and 19.24% of cleared value; no clearing fails D5. Appendix D gives the same shares per defect class, with the clustered intervals, per item category, spend area, month, line value quartile and vendor activity quartile, reported beside the estimates and never modelled in. Section 4 reports the registered category cross-check, the cleared-value sources and the exported sample.

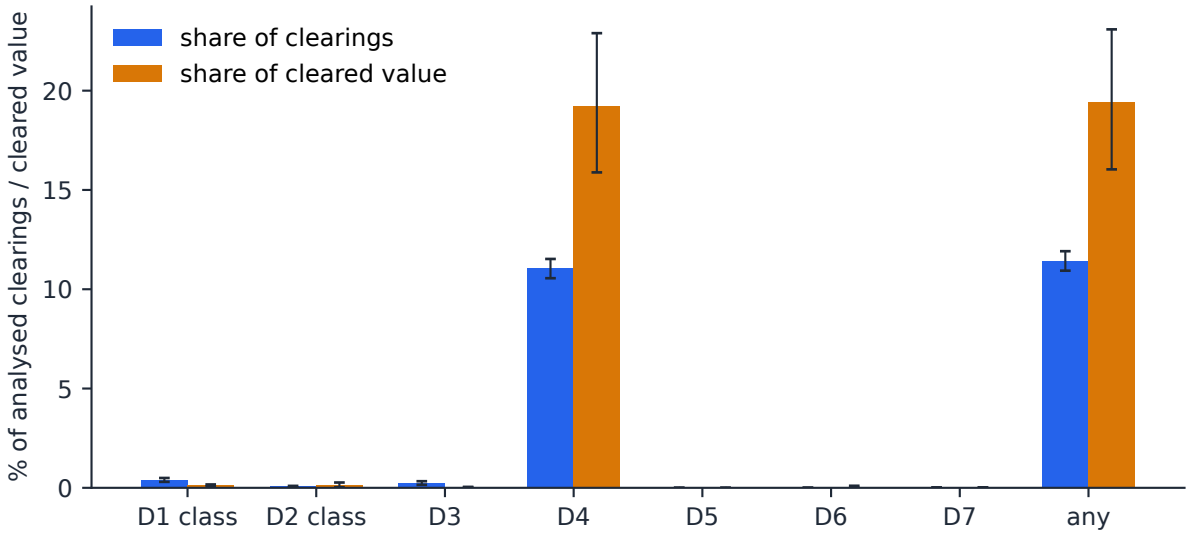



Figure 1: share of analysed clearings and of cleared value per defect class, with clustered bootstrap 95% intervals.

**D4 sensitivity (exploratory).** The registered reading of D4 counts the creation as approval-bearing (Section 3). Of the 17,157 D4 failures, 16,478 have no explicit approval or release event before the clearing; 679 clearings fail D4 after an explicit approval or release event. Exposure under the three readings:

| reading (exploratory) | clearings | share | Wilson 95% | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| (a) registered: at least one failed predicate | 17,748 of 155,534 | 11.41% | [11.25%, 11.57%] | [10.94%, 11.92%] | 19.41% | [16.04%, 23.09%] |
| (b) D4 counts only changes after an explicit approval or release event | 1,405 of 155,534 | 0.90% | [0.86%, 0.95%] | [0.67%, 1.19%] | 5.65% | [3.67%, 8.13%] |
| (c) D4 failures with the creation as the only approval-bearing event | 16,478 of 155,534 | 10.59% | [10.44%, 10.75%] | [10.17%, 11.04%] | 13.79% | [11.49%, 16.49%] |

Source: `results/exploratory/d4-sensitivity.json` (exploratory).

Under the strict reading, H2's O rates at W = 60 days are 4.42% against 1.05% (difference 0.0337, clustered 95% [0.0129, 0.0736]), beside the registered difference 0.0094 (Section 5.2); exploratory, no p-value. Materiality: the `Cumulative net worth (EUR)` value is repeated on every event of a line, so the magnitude of a counted change is observable on 13 of 17,157 D4 failures only. Restricting D4 failures to observable relative changes above the unregistered thresholds 0, 1%, 5% and 10% gives the same 741 clearings (0.48%) at every threshold: the clearings that fail a predicate other than D4. The magnitude of a change is therefore an undetected class (Section 6).

### 5.2 Consequence (E2, registered)

| test | defect group (failed predicate): O rate | comparison: O rate | difference | bootstrap SE | z | p (raw, one-sided) | p (Holm) |
|---|---|---|---|---|---|---|---|
| H1 | 39 of 692 (5.64%) | 1,286 of 137,602 (0.93%) | 0.0470 | 0.0127 | 3.688 | $1.13 \times 10^{-4}$ | $4.53 \times 10^{-4}$ |
| H2 | 327 of 17,157 (1.91%) | 1,332 of 138,377 (0.96%) | 0.0094 | 0.0018 | 5.266 | $6.96 \times 10^{-8}$ | $3.48 \times 10^{-7}$ |
| H3 | 0 of 0 (n/a) | 1,659 of 155,534 (1.07%) | n/a | n/a | n/a | not computable | 1 |

O is the registered corrective within W = 60 days. The H1 comparison group is three-way clearings with every applicable predicate passing; against clearings of every category with every applicable predicate passing the difference is 0.0470, descriptively. H2 compares D4 fail with D4 pass in every category. Standard errors come from the purchase-document-clustered bootstrap. Holm's adjustment runs over the registered family of five tests, H1, H2, H3 and H4 for each model family, at alpha = 0.05:

| test | p (raw) | rank | multiplier | p (Holm) | below alpha |
|---|---|---|---|---|---|
| H2 | $6.96 \times 10^{-8}$ | 1 | 5 | $3.48 \times 10^{-7}$ | yes |
| H1 | $1.13 \times 10^{-4}$ | 2 | 4 | $4.53 \times 10^{-4}$ | yes |
| H3 | 1 | 3 | 3 | 1 | no |
| H4-Haiku | 1 | 4 | 2 | 1 | no |
| H4-Jev | 1 | 5 | 1 | 1 | no |

H3, distinct actors, is not computable: defect group has no analysed clearings. No analysed clearing was made by an actor of its line's earliest creation event, so the defect class has zero incidence; across the log the actors who record a `Clear Invoice` and those who record the earliest creation of a line are separate populations (23 and 185 actors, 0 in common; Section 6). No analysed clearing has D5 unknown (0): the 7,941 `Clear Invoice` events with no actor all lie outside the analysed set: of the 94 on framed lines, 94 fall after the horizon T_end - W (`results/e1/analysed-clearing-actors.json`). H3 enters the family with p = 1, the convention the author fixed in decision D-C-2, committed at 2026-10-25 18:21:56 UTC, before the first model request at 2026-10-25 18:42:18 UTC (`results/e4/timeline.json`). At W = 30 days the differences are 0.0354 for H1 and 0.0055 for H2, descriptively. Appendix E gives H1 and H2 per item category, spend area, value quartile, vendor activity quartile and month, descriptively and never modelled in.

**Standardised H1 and H2 (exploratory).** Each registered difference beside the difference standardised to the comparison group's stratum distribution, for three stratifications; strata with no clearings in the failing group are excluded and their weight in the comparison group is shown. No regression, no p-value:

| test | estimate | difference | clustered 95% | excluded comparison weight |
|---|---|---|---|---|
| H1 | registered, unadjusted | 0.0470 | [0.0246, 0.0747] | |
| H1 | exploratory, standardised by category x line-value quartile | 0.0662 | [0.0378, 0.0984] | 5.85% |
| H1 | exploratory, standardised by month of t_c | 0.0507 | [0.0290, 0.0824] | 0.00% |
| H1 | exploratory, standardised by spend area | 0.0778 | [0.0461, 0.1125] | 3.35% |
| H2 | registered, unadjusted | 0.0094 | [0.0061, 0.0130] | |
| H2 | exploratory, standardised by category x line-value quartile | 0.0031 | [0.0008, 0.0054] | 0.06% |
| H2 | exploratory, standardised by month of t_c | 0.0094 | [0.0060, 0.0132] | 0.00% |
| H2 | exploratory, standardised by spend area | 0.0103 | [0.0070, 0.0137] | 0.00% |

Sources: `results/e2/hypotheses.json` (registered), `results/exploratory/adjusted-h1-h2.json` (exploratory).

H1 holds under every stratification; H2 holds, by category and value quartile at about a third of its unadjusted size.

### 5.3 Gate effect and hold cost (E3, registered)

The gate would have held 17,748 of 155,534 analysed clearings (11.41%, 19.41% of cleared value); 0 of them were held on an unknown predicate value alone. Against the corrective at W = 60 days, in clearings and anonymised EUR (Figure 2):

| gate verdict | corrective within W | no corrective within W |
|---|---|---|
| would have held | 368 (21,418,557) | 17,380 (145,068,897) |
| would have allowed | 1,291 (106,294,821) | 136,495 (584,977,346) |

Appendix E gives the same table per stratum.

**Hold friction (exploratory).** The share of the gate's holds followed by no corrective within W, in count and in cleared value:

| window | gate holds | without a corrective | clustered 95% | share of held value | clustered 95% |
|---|---|---|---|---|---|
| W = 30 days | 17,748 | 98.65% | [98.37%, 98.90%] | 92.07% | [86.68%, 96.07%] |
| W = 60 days | 17,748 | 97.93% | [97.57%, 98.26%] | 87.14% | [81.88%, 91.93%] |

Source: `results/exploratory/hold-friction.json` (exploratory).

A hold with no corrective within W is not shown to be a false hold: the proxy is what it is. The share is the review load the contract would create, stated as a proportion and nothing more.

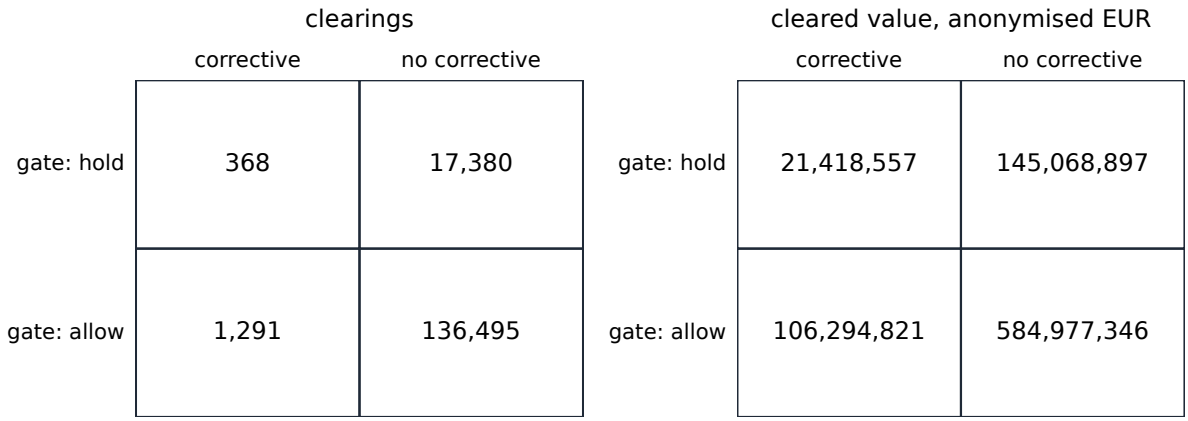



Figure 2: gate verdict against the corrective within W = 60 days, in clearings and in cleared value.

For D1/D2 failures, the time from clearing to the first uncancelled receipt (Figure 3):

| lag band | clearings | cleared value |
|---|---|---|
| [0, 1] days | 11 | 109,886 |
| (1, 7] days | 21 | 113,119 |
| (7, 30] days | 11 | 803,512 |
| over 30 days | 39 | 44,240 |
| never (by T_end) | 610 | 1,015,565 |

**Human hold (registered).** Of 180,664 framed lines, people held 53,262 and released 101,777; 25,625 had no invoice receipt by T_end - W and are not classified. The status of a line is the gate verdict at the decision point t* of the agent replay: the first clearing if it came within W of the first invoice receipt, otherwise that receipt plus W. The registered text left D5 at a t* that is not a clearing undefined; read literally it is unknown, and then a line with no clearing within W can never be clean under the literal reading. The author amended the reading before any model call (at a t* that is not a clearing, D5 is not applicable: the decider there is the gate or a model; decision D-C-2 B); both readings are reported:

| reading | predicate status at t* | classified lines | held by people | human hold rate |
|---|---|---|---|---|
| literal (registered text) | defective | 16,264 | 5,664 | 34.83% |
| literal (registered text) | clean | 91,178 | 1 | 0.00% |
| literal (registered text) | held unknown | 47,597 | 47,597 | 100.00% |

| reading | predicate status at t* | classified lines | held by people | human hold rate |
|---|---|---|---|---|
| amended (D-C-2 B) | defective | 16,264 | 5,664 | 34.83% |
| amended (D-C-2 B) | clean | 138,775 | 47,598 | 34.30% |
| amended (D-C-2 B) | held unknown | 0 | 0 | n/a |

Under the amended reading, people held 34.83% of lines with a failed predicate at t* and 34.30% of clean lines.

### 5.4 People, models and the gate on the same lines (E4, registered)

**Measurement asymmetry.** The models and the gate decide at t*; the people column is observed non-release within W. Of the 53,262 framed lines held by people, 53,182 are non-clearance within W alone, 10 a payment block alone and 70 both (`results/e3/human-hold-composition.json`). The tables name the column accordingly.

**Sample and arms.** From the framed lines with an invoice receipt, 18,006 had at least one failed predicate at t* and 147,240 were clean, under D-C-2 B (`results/e4/sample-frame.json`). These differ from the 16,264 and 138,775 lines of Section 5.3 (`results/e3/human-hold.json`), which are the lines the people arm can classify: a line whose first invoice receipt comes later than T_end - W is eligible for the sample but not classified (section 3.6). The registered sample drew 1,000 defective and 1,000 clean lines, at most one per purchase document, stratified by category and value quartile, with seed 20261201; the draw was committed before any call. Each line was shown to two model families as the registered ERP view at t*, a short record with no event timeline and no company, vendor, actor or line id: TypeSafe System One (Jev, returned model `jev-1.14.0`) through `typesafe-sdk==0.7.2`, asked one choice question with labels clear and hold, and the Anthropic Messages API (`claude-haiku-4-5-20251001`, temperature zero, assistant prefill), asked for a JSON decision and hold probability. For Jev the template's closing reply-format sentence was replaced by that typed choice, a deviation (Section 4); Haiku received the template as registered. Invalid replies, after one re-request: 0 for Jev and 0 for Haiku. The people arm is read from the log (Section 3); 152 sample lines have no invoice receipt by T_end - W and are not classified (section 3.6), so its denominators are smaller. The gate's verdict holds every line with a failed predicate and allows every clean line by construction.

| arm | held, lines with a failed predicate | held, clean lines | kappa with gate | value released on lines with a failed predicate | expected correctives |
|---|---|---|---|---|---|
| people (observed non-release within W) | 301 of 912 (33.0%) | 318 of 936 (34.0%) | n/a | 4,759,520 | 12.260 |
| Jev | 1,000 of 1,000 (100.0%) | 530 of 1,000 (53.0%) | 0.470 | 0 | 0.000 |
| Haiku | 996 of 1,000 (99.6%) | 573 of 1,000 (57.3%) | 0.423 | 7,307 | 0.076 |
| gate | 1,000 of 1,000 (100.0%) | 0 of 1,000 (0.0%) | n/a | 0 | 0.000 |

| family | d (failed predicate minus clean) | SE | p, lower | p, upper | p_TOST |
|---|---|---|---|---|---|
| H4-Jev | 0.470 | 0.0158 | $2.33 \times 10^{-238}$ | 1 | 1 |
| H4-Haiku | 0.423 | 0.0158 | $5.54 \times 10^{-198}$ | 1 | 1 |

Source: `results/e4/arms.json`, `results/e4/h4.json`.

People: observed non-release within W (53,182 non-clearance alone, 10 payment block alone, 70 both, among the framed lines held by people; Section 5.3).

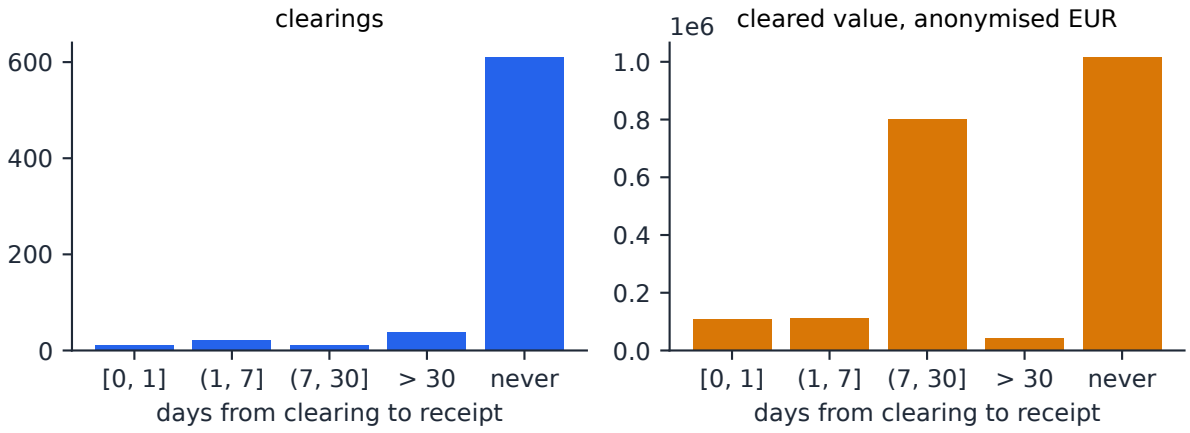



Figure 3: D1/D2 failures by lag band, in clearings and cleared value.

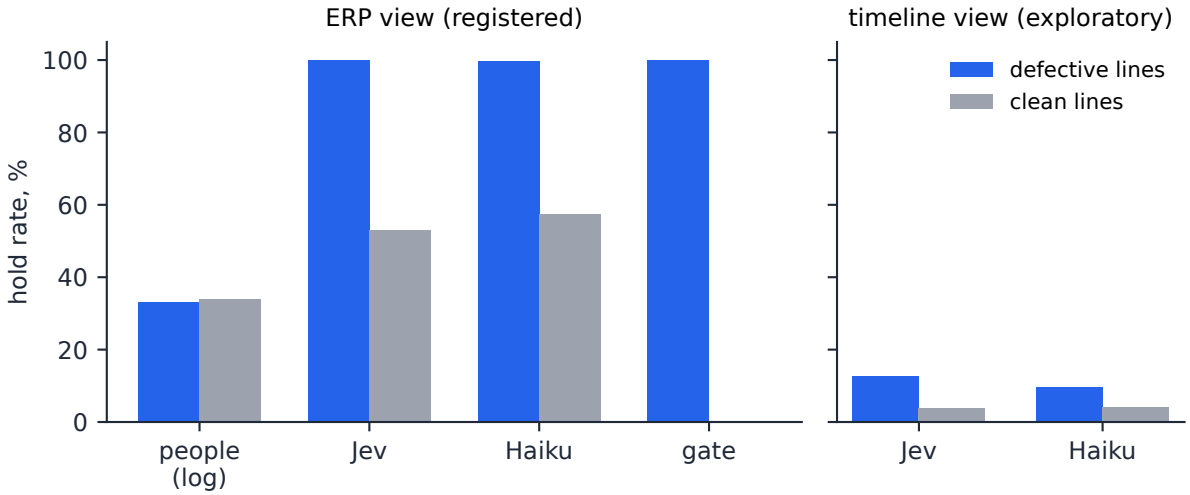



Figure 4: hold rate on lines with a failed predicate and on clean lines, per arm; the exploratory timeline arm in its own panel.

H4 asks whether each family's hold rate on defective lines (a failed predicate) is within m = 0.05 of its rate on clean lines. Neither is: Jev held 100.0% and 53.0%, Haiku held 99.6% and 57.3%; p_TOST is 1 for Jev and 1 for Haiku, and H4 is below alpha after Holm for neither. Both families track the gate's verdict on the lines the gate would have held and hold about half of the lines it allows (Cohen's kappa with the gate 0.470 and 0.423). Released on lines with a failed predicate, in anonymised EUR: people 4,759,520, Jev 0, Haiku 7,307, gate 0; for the gate, held lines go somewhere the log does not show. The expected correctives are the metric of registration section 6 (E4 row): each released line with a failed predicate carries the E2 O rate at W = 60 days of its defect group. The precedence and the unmatched-group rule are decision D-C-3: the H1 group if D1/D2 fails and otherwise the H2 group if D4 fails; a line failing neither (only D3, D6 or D7) has no E2 group and is counted, not imputed (8 sample lines, 1 of them released by people). The ERP view hands each model the inputs of the predicates; this result is about the view as much as about the models. The expected-corrective totals are the number of corrective events the E2 rates would predict for the lines each arm released, under the stated assignment: a projection from observed rates onto the sample, not an observed count.

**On the lines the people arm classifies (exploratory re-tabulation).** The same four arms, re-tabulated on the 1,848 sample lines the people arm classifies; the table above, on all sample lines, stays primary:

| arm | held, lines with a failed predicate | held, clean lines | kappa with gate | value released on lines with a failed predicate | expected correctives |
|---|---|---|---|---|---|
| people (observed non-release within W) | 301 of 912 (33.0%) | 318 of 936 (34.0%) | n/a | 4,759,520 | 12.260 |
| Jev | 912 of 912 (100.0%) | 491 of 936 (52.5%) | 0.472 | 0 | 0.000 |
| Haiku | 908 of 912 (99.6%) | 524 of 936 (56.0%) | 0.433 | 7,307 | 0.076 |
| gate | 912 of 912 (100.0%) | 0 of 936 (0.0%) | n/a | 0 | 0.000 |

Source: `results/e4/arms-classified-lines.json`; a re-tabulation on the lines the people arm classifies.

People: observed non-release within W (53,182 non-clearance alone, 10 payment block alone, 70 both, among the framed lines held by people; Section 5.3).

**What the people arm measures.** The log records no hold decision except a payment block. "Held by people" (section 3.6) is a payment block before the first clearing, or no clearing within W of the first invoice receipt: an inferred non-release, not the same act as a model's hold verdict at t*. Of the 53,262 framed lines held by people, 10 were held by a payment block alone, 53,182 by no clearing within W alone, and 70 by both:

| reading | predicate status at t* | held by people | payment block before the first clearing | no clearing within W of the first invoice | both |
|---|---|---|---|---|---|
| literal (registered text) | defective | 5,664 | 9 | 5,619 | 36 |
| literal (registered text) | clean | 1 | 1 | 0 | 0 |
| literal (registered text) | held unknown | 47,597 | 0 | 47,563 | 34 |
| amended (D-C-2 B) | defective | 5,664 | 9 | 5,619 | 36 |
| amended (D-C-2 B) | clean | 47,598 | 1 | 47,563 | 34 |
| amended (D-C-2 B) | held unknown | 0 | 0 | 0 | 0 |

Source: `results/e3/human-hold-composition.json`.

**Exploratory: the event timeline instead of the ERP view.** Labelled exploratory and outside the Holm family: the same lines, the same families and rules, with the line's events up to t* (date, activity, value) in place of the ERP view.

| family (timeline view, exploratory) | held, lines with a failed predicate | held, clean lines | kappa with gate | invalid replies |
|---|---|---|---|---|
| Jev | 126 of 1,000 (12.6%) | 38 of 1,000 (3.8%) | 0.088 | 0 |
| Haiku | 95 of 1,000 (9.5%) | 39 of 1,000 (3.9%) | 0.056 | 0 |

Source: `results/e4-exploratory/arms-exploratory.json`; exploratory, no interval, outside the Holm family.

Under the timeline view the same models held far fewer lines with a failed predicate (exploratory, no interval). The experiment does not separate instruction wording, field structure and information content: the timeline view was sent with the template's opening paragraph, which says "Using only the ERP record below", followed by the timeline, as Appendix F shows. The gate's column follows from the predicates by construction: it is exact contract enforcement, not decision accuracy. Spend for both families and both views together was USD 2.34 against the registered cap of USD 60 (`results/e6/cost.json`).

### 5.5 Derivation (E5, registered)

Paper 5's compiler, at the pinned commit and unmodified, was given per category the observed states of the evidence set and the three declared losses of the registration: L1, pay for goods not received (the category requires a goods receipt and none is uncancelled); L2, pay twice (the clearing's position among the line's clearings exceeds the invoices recorded); L3, pay a changed amount (a change since approval). The derived contract is the set of registered predicates whose observation is in the compiler's minimum-cardinality sufficient set. The comparison is the registered one (registration section 6, E5): "whether the derived contract equals the category contract in 3.1".

| category | clearings | states | L1 | L2 | L3 | fewest fields | derived contract | registered contract | result |
|---|---|---|---|---|---|---|---|---|---|
| 2-way | 188 | 3 | 0 | 0 | 4 | `changes` | D4 | D3, D4, D5, D6, D7 | does not reproduce |
| 3-way, invoice after receipt | 9,640 | 896 | 0 | 81 | 1,585 | `changes`, `clearing_ordinal`, `invoices` | D3, D4 | D1, D2, D3, D4, D5, D6, D7 | does not reproduce |
| 3-way, invoice before receipt | 145,706 | 758 | 692 | 1,390 | 15,568 | `changes`, `clearing_ordinal`, `invoices`, `receipts` | D1, D2, D3, D4 | D1, D2, D3, D4, D5, D6, D7 | does not reproduce |

The compiler does not reproduce the registered contract in any category. The three losses derive at most D1 to D4: no declared loss reads the observations of D5, D6 or D7, so those predicates correspond to no declared loss. In the before-receipt category the derived D1 to D4 is the three-way match in the narrow sense (receipt, invoice and an unchanged approved amount), and the registered contract carries three predicates beyond it. In the after-receipt category the compiler derives D3 and D4 only, because L1 occurs on 0 analysed clearings there, and the registered contract carries five predicates beyond it. This is a limit of deriving from one log: a loss the log never shows calls for no observation. Pay twice needs the clearing's position on its line, an observation no registered predicate reads.

Read the other way, paper 5's sufficiency function applied to the registered contract over the same states (`results/e5/sufficiency.json`):

| category | registered fields | sufficient for | not sufficient for | clearings with L2 | minimal | derived contract a subset |
|---|---|---|---|---|---|---|
| 2-way | `invoices`, `changes`, `distinct_actors`, `blocks`, `closed` | L1, L2, L3 | none | 0 | no | yes |
| 3-way, invoice after receipt | `receipts`, `invoices`, `changes`, `distinct_actors`, `blocks`, `closed` | L1, L3 | L2 | 81 | no | yes |
| 3-way, invoice before receipt | `receipts`, `invoices`, `changes`, `distinct_actors`, `blocks`, `closed` | L1, L3 | L2 | 1,390 | no | yes |

The registered contract is sufficient, not derived: sufficient for the three declared losses in the two-way category, and in the three-way categories for L1 and L3 but not for L2, for the reason above; minimal in no category (C1). The compiler's contract is smaller in every category, and so C9, which asks whether it equals the registered one, is not supported.

**What the negative result shows (discussion of the registered result).** For each predicate the compiler did not derive, the reason is one of two. D5, D6 and D7 read observations that no declared loss reads: the reason is a loss not declared. Pay twice (L2) needs an observable the registered contract does not carry, the clearing's position on its line. A minimum-cardinality sufficient contract over the observed states is not the normative contract: a loss that never occurs in the history yields no predicate (L1 in the after-receipt category). For paper 5's method to carry such controls, each needs a declared loss and a declared observable, not history.

### 5.6 Governance cost (E6, registered)

The full replay of every line, from reading the log to the written evidence set, took a median of 153.5 seconds over three runs on a 4-core machine, 0.000987 seconds per gate decision. A model decision cost USD 0.000022 with Jev and USD 0.000368 with Haiku.

### 5.7 Applicability to another log: travel declarations (E7, secondary)

E7 tests whether the procedure applies to another log, not external validity. An informative replication needs recorded approvals, explicit decision events, reliable actor identities, enough failed predicates, and a downstream outcome within the window.

The unit is a payment on a declaration case at least W before the latest timestamp of its log; D1' approval present, D2' approval by the required role and D3' unchanged after approval are evaluated before the payment, and O' is a rejection or resubmission within W after it. Descriptive and secondary, outside the Holm family.

| log | cases | payments | excluded after T_end' - W | units |
|---|---|---|---|---|
| domestic | 10,500 | 10,044 | 2 | 10,042 |
| international | 6,449 | 6,187 | 0 | 6,187 |

| scope | predicate | failing payments | O' rate, failing | O' rate, all-pass | difference | bootstrap 95% |
|---|---|---|---|---|---|---|
| domestic | D1' | 2 | 0.500 | 0.000 | 0.500 | [0.0000, 1.0000] |
| domestic | D2' | 2 | 0.500 | 0.000 | 0.500 | [0.0000, 1.0000] |
| domestic | D3' | 0 | n/a | 0.000 | n/a | n/a |

| scope | predicate | failing payments | O' rate, failing | O' rate, all-pass | difference | bootstrap 95% |
|---|---|---|---|---|---|---|
| international | D1' | 0 | n/a | 0.000 | n/a | n/a |
| international | D2' | 0 | n/a | 0.000 | n/a | n/a |
| international | D3' | 0 | n/a | 0.000 | n/a | n/a |
| pooled | D1' | 2 | 0.500 | 0.000 | 0.500 | [0.0000, 1.0000] |
| pooled | D2' | 2 | 0.500 | 0.000 | 0.500 | [0.0000, 1.0000] |
| pooled | D3' | 0 | n/a | 0.000 | n/a | n/a |

The replication is inconclusive. Only 2 of 16,229 payments fail any predicate; the 2 domestic payments with D3' unknown are the same 2 that fail D1' and D2' (`results/e7/unknown-overlap.json`). O' follows 0 of the 16,227 payments with every predicate passing; no comparison has an interval excluding zero. A bootstrap replicate in which either group is empty is discarded and the percentiles are taken over the retained replicates (8,685 of 10,000 for the domestic log and 8,696 pooled, `results/e7/comparisons.json`; a deviation, Section 4). The registered names `Declaration FINAL_APPROVED by DIRECTOR` and `Declaration REJECTED by DIRECTOR` do not occur in the domestic log; they are reported with zero counts, not replaced.

## 6 Coverage

What the replay can and cannot see, as registered:

**Detected (BPI 2019, E1 to E6)**

| defect class | predicate | dimension | detected from |
|---|---|---|---|
| goods receipt required but absent at clearing | D1 receipt_present | presence | receipts minus cancellations at most zero before t_c (count of `Record Goods Receipt` minus count of `Cancel Goods Receipt`), on a line whose category requires receipt |
| stale receipt: receipt recorded after clearing | D2 receipt_before_clearing | freshness, order | the first uncancelled `Record Goods Receipt` falls after t_c (lag band registered) |
| invoice absent at clearing | D3 invoice_present | presence | no `Record Invoice Receipt` before t_c |
| changed after approval | D4 unchanged_since_approval | validity | `Change Price` or `Change Quantity` between the last approval-bearing event and t_c |
| same actor created and cleared | D5 distinct_actors | authority | actor of `Create Purchase Order Item` equals actor of `Clear Invoice` |
| clearing under an open payment block | D6 block_respected | scope | count of `Set Payment Block` minus count of `Remove Payment Block` before t_c greater than zero |
| line deleted or blocked at clearing | D7 not_deleted | validity | count of `Delete Purchase Order Item` plus `Block Purchase Order Item` minus `Reactivate Purchase Order Item` before t_c greater than zero |

**Detected (BPI 2020 declarations, E7, secondary)**

| defect class | predicate | dimension | detected from |
|---|---|---|---|
| paid without an approval event | D1' approval_present | presence | no event in the registered approval list before the payment |
| approved by a role other than the required one | D2' approval_by_required_role | authority | the approval chain before payment lacks the registered required role |
| declaration changed after approval | D3' unchanged_since_approval | validity | a resubmission by the employee between the last approval and payment |

### Undetected, with reason

| defect class | reason |
|---|---|
| wrong amounts in the payload | values are anonymised by a linear translation; the log carries cumulative net worth per event, not invoice line content, so a mismatch between invoice text and goods is not observable |
| fraud | intent is not an event attribute; no predicate over the log separates a fraudulent clearing from an erroneous one, and the paper makes no statement about it |
| vendor identity (wrong, duplicate or fictitious vendor) | vendor names and ids are anonymised consistently; the log carries no master-data change history or bank details |
| recovery of corrected payments | the log records the corrective event and nothing about recovery; whether money was returned, offset or written off is not recorded |
| what a held line becomes | a gate hold is a replay verdict; the log records the human path only, so the fate of a line the gate would have held is not observable |
| magnitude of a price or quantity change | the value field is cumulative and repeated on every event of a line, so the size of a counted change is observable on 13 of 17,157 D4 failures (`results/exploratory/d4-sensitivity.json`) |
| approvals outside the log | approvals given by e-mail, in another system or verbally leave no event; D4's "last approval-bearing event" sees only logged approvals |
| partial receipt and value mismatch per receipt | the replay checks presence and order of the receipt, not the matched values; values are anonymised and the registered mapping reads no per-receipt quantity |
| duplicate invoice across lines | the case is one purchase-order line and the registered mapping carries no invoice document id, so an invoice cleared on two lines is not linked |

Per class, the failures found (BPI 2019 on analysed clearings, BPI 2020 on payments):

| log | predicate | failed | unknown |
|---|---|---|---|
| BPI 2019 (analysed clearings) | D1 | 692 | 0 |
| BPI 2019 (analysed clearings) | D2 | 692 | 0 |
| BPI 2019 (analysed clearings) | D3 | 362 | 0 |
| BPI 2019 (analysed clearings) | D4 | 17,157 | 0 |
| BPI 2019 (analysed clearings) | D5 | 0 | 0 |
| BPI 2019 (analysed clearings) | D6 | 4 | 0 |
| BPI 2019 (analysed clearings) | D7 | 19 | 0 |
| BPI 2020 domestic (payments) | D1' | 2 | 0 |
| BPI 2020 domestic (payments) | D2' | 2 | 0 |
| BPI 2020 domestic (payments) | D3' | 0 | 2 |
| BPI 2020 international (payments) | D1' | 0 | 0 |
| BPI 2020 international (payments) | D2' | 0 | 0 |
| BPI 2020 international (payments) | D3' | 0 | 0 |

Sources: `results/e1/coverage-check.json`, `results/e1/predicate-unknowns.json` (BPI 2019 unknown), `results/e7/predicates.json`.

On BPI 2019 every detected class fails at least once except the authority class (D5), with 0 failed clearings: across the log, the 23 actors who record a `Clear Invoice` and the 185 who record a line's earliest creation share 0, so the two are separate populations; no `Clear Invoice` is recorded by a batch user (an actor whose name starts with the registered batch prefix), and 7,941 carry no actor (`results/e1/actors.json`), none of them on an analysed clearing (0; Section 5.2). D5 fires on synthetic lines in the engine's tests (`test_d5_actors` and `test_verdict_at_decision_point_d5_off_clearing` in `tests/test_engine_units.py`, and fixture line C in `tests/test_pipeline.py`). On BPI 2020, D3' fails on no payment in either log, and in the international log no predicate fails at all.

## 7 Limitations and non-claims

The outcome is a proxy: a corrective event after a clearing is not proof that the clearing was wrong, and the study is observational. Category, spend area, value, vendor activity and month are reported beside every estimate (Appendix D for exposure, Appendix E for H1, H2 and the gate by outcome) and never modelled in. Values are anonymised. What a line the gate would have held becomes is not in the log. The people column reports what the process did within W; the comparison across the people, the models and the gate is of outcomes of three deciders, not of three acts. The only inference the three columns support is this: on the 1,848 sample lines the log classifies, the process released about two thirds of both sets within W (people held 33.0% and 34.0%); the models shown the predicate inputs held nearly every line with a failed predicate (100.0% and 99.6%) and about half of the clean ones (52.5% and 56.0%; 53.0% and 57.3% on all 2,000 sample lines); the gate held exactly the lines with a failed predicate. No decision quality is inferred for people, whose column is a process outcome. The ERP view hands the model the inputs of the predicates; under the timeline view the same models held far fewer lines (exploratory, no interval), which is a statement about views, not about the models alone. The derivation depends on how the three declared losses are expressed over the evidence set; the expressions are recorded with the result. One company's process, one year.

## 8 Conclusion

11.41% of the 155,534 analysed invoice clearings in this process were made on evidence that failed at least one registered predicate, carrying 19.41% of the cleared value; those made with an absent or late receipt (D1/D2) were followed by a corrective at a higher rate than three-way clearings with every applicable predicate passing, and those made after a change since approval (D4 fail) at a higher rate than D4-pass clearings. Counting only changes after an explicit approval, an exploratory reading, exposure is 0.90% of clearings and 5.65% of cleared value, beside the registered 11.41% and 19.41%. A failed registered predicate is not a confirmed violation of the company's approval policy; whether a change after creation required re-approval is not recorded in the log. The sufficiency check adds that the registered contract is sufficient for "goods not received" and "changed amount", and not for "pay twice" in the three-way categories, because no registered predicate reads the clearing's position on its line.

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## Appendix A: Registration summary

The registered numbers, Section 9 of the registration:

| item | value |
|---|---|
| window | Create Purchase Order Item in calendar 2018 (UTC) |
| W | 60 days; sensitivity 30 days |
| lag bands | [0, 1], (1, 7], (7, 30], over 30 days, never |
| K (framework exclusion) | more than 12 clearings |
| E0 | 500 lines |
| E4 | N_d = 1,000, N_c = 1,000; m = 0.05 |
| alpha | 0.05, Holm over H1, H2, H3, H4-Jev, H4-Haiku |
| bootstrap | B = 10,000, purchase-document clusters |
| seed | 20261201 |
| cap | USD 60 |
| pins | `engines.lock` at this commit |

## Appendix B: Claims, decisions and deviations

Claim statuses, each set from its slot file:

| claim | status | evidence |
|---|---|---|
| C1 | supported | `results/e5/sufficiency.json`, `results/e1/evidence-set.jsonl.gz`, `tests/test_properties.py`; engine evidence `results/mutation.json` |
| C2 | supported | `results/e1/exposure.json`, `results/e1/exposure-by-stratum.json` |
| C3 | supported | `results/e2/hypotheses.json` |
| C4 | supported | `results/e2/hypotheses.json` |
| C5 | inconclusive: defect class has zero incidence in the analysed clearings | `results/e1/coverage-check.json`, `results/e2/hypotheses.json` |
| C6 | not supported for both families | `results/e4/h4.json` |
| C7 | supported | `results/e3/gate-held.json`, `results/e3/verdict-by-outcome.json`, `results/e3/lag-bands.json` |
| C8 | supported | `results/e4/arms.json` |
| C9 | not supported: 2-way does not reproduce; 3-way, invoice after receipt does not reproduce; 3-way, invoice before receipt does not reproduce | `results/e5/derivation.json` |
| C10 | supported | `results/e6/timing-e1.json`, `results/e6/cost.json` |
| C11 | inconclusive | `results/e7/comparisons.json` |
| C12 | supported; D5 class zero incidence | `results/e1/coverage-check.json` |
| C13 | supported | `results/reproducibility.json`, `results/e4/double-run.json`, `results/reproduction/attempt-3.json`, issue https://github.com/besanson/sarc-p2p-replay/issues/1 |

Reading notes, as in the claims file. C1's "(paper 5)" refers to paper 5's contract construct, whose sufficiency for the declared losses over the observed states E5 checks with paper 5's own function, per category and per loss (`results/e5/sufficiency.json`): sufficient for every declared loss in the two-way category, and for L1 and L3 but not L2 in the three-way categories; minimal in none; each predicate checks one SARC-DQ dimension (Section 3). C2's "relied on" is read as "made on evidence that failed". C9's "per-category matching rule" is the category contract of section 3.1, the comparator the registration's E5 row fixes; the before-receipt category recovers the narrow three-way match (D1 to D4), and the status "not supported" refers to the registered comparison (Section 5.5). C2's exposure is the registered predicate set; its D4 component depends on the registered reading of the creation as approval-bearing; the strict reading is reported as exploratory (Section 5.1).

Every decision taken while the work ran is in the repository's decision log, from D-A0-1 to D-G-2, and every deviation from the tagged registration is in its deviations file (Section 4).

## Appendix C: Reproduction

`make bootstrap` clones the sibling repositories at their pins, creates the environment, fetches the logs and checks their hashes; `make test` and `make lint` run the tests and the lints; `make gates` runs the release gates G1 to G9; the experiment runners regenerate every slot under `results/`, and `make paper` regenerates this paper and its figures from the slots. Internal double runs on one machine are byte-identical (16 files for E1 to E3). Across machines and architectures (x86-64 Linux at the builder; arm64 macOS in the first external attempt, issue #1), every result file is equal to within 1e-12 relative; the last floating-point digits can differ (maximum observed absolute difference 1.1e-16); figure bytes differ by font embedding; the paper text, its TeX source and the README regenerate byte-identically. External reproduction: Eduardo Arana, Independent Researcher, on 2026-11-12, in a clean checkout of the release (the clone of attempt 1) on macOS arm64 (issue #1, attempt 3): 51 files equal, 46 byte-identical, 5 within tolerance, maximum difference 1.11e-16; attempts 1 and 2 failed on tooling errors corrected in 1.0.1 and 1.0.2 and are recorded. The reproduced release is V1.0.2, whose registered results 1.0.3 and 1.1.0 carry unchanged; the exploratory slots added in 1.1.0 are regenerated by the same `make reproduce` and were byte-identical in the fresh-clone check of 1.1.0.

## Appendix D: Exposure by stratum

The share of analysed clearings and of cleared value per defect class (a failed predicate), per stratum, with the purchase-document-clustered 95% intervals; classes with no clearing in a stratum are omitted here and kept in `results/e1/exposure-by-stratum.json`. Company and vendor strata are in the slot file only, as hashed labels. The spend-area labels are the public log's own strings and the registered strata; the challenge itself names the sector.

**Item category**

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| 2-way | D4 | 4 of 188 | 2.13% | [0.00%, 4.94%] | 2.68% | [0.00%, 8.71%] |
| 2-way | at least one defect | 4 of 188 | 2.13% | [0.00%, 4.94%] | 2.68% | [0.00%, 8.71%] |
| 3-way, invoice after receipt | D3 | 9 of 9,640 | 0.09% | [0.01%, 0.21%] | 0.02% | [0.00%, 0.05%] |
| 3-way, invoice after receipt | D4 | 1,585 of 9,640 | 16.44% | [14.36%, 18.73%] | 15.65% | [9.48%, 24.64%] |
| 3-way, invoice after receipt | at least one defect | 1,592 of 9,640 | 16.51% | [14.42%, 18.79%] | 15.67% | [9.50%, 24.66%] |
| 3-way, invoice before receipt | D1 class (absent receipt) | 610 of 145,706 | 0.42% | [0.32%, 0.53%] | 0.21% | [0.14%, 0.28%] |
| 3-way, invoice before receipt | D2 class (stale receipt) | 82 of 145,706 | 0.06% | [0.02%, 0.10%] | 0.22% | [0.05%, 0.45%] |
| 3-way, invoice before receipt | D3 | 353 of 145,706 | 0.24% | [0.15%, 0.35%] | 0.04% | [0.02%, 0.06%] |
| 3-way, invoice before receipt | D4 | 15,568 of 145,706 | 10.68% | [10.21%, 11.19%] | 21.92% | [20.71%, 23.18%] |
| 3-way, invoice before receipt | D6 | 4 of 145,706 | 0.00% | [0.00%, 0.01%] | 0.05% | [0.00%, 0.16%] |
| 3-way, invoice before receipt | D7 | 19 of 145,706 | 0.01% | [0.01%, 0.02%] | 0.02% | [0.00%, 0.04%] |
| 3-way, invoice before receipt | at least one defect | 16,152 of 145,706 | 11.09% | [10.60%, 11.60%] | 22.21% | [21.00%, 23.48%] |

**Spend area**

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Additives | D1 class (absent receipt) | 9 of 10,206 | 0.09% | [0.04%, 0.15%] | 0.13% | [0.02%, 0.29%] |
| Additives | D2 class (stale receipt) | 4 of 10,206 | 0.04% | [0.01%, 0.08%] | 0.02% | [0.00%, 0.04%] |
| Additives | D3 | 4 of 10,206 | 0.04% | [0.01%, 0.08%] | 0.01% | [0.00%, 0.04%] |
| Additives | D4 | 1,207 of 10,206 | 11.83% | [11.10%, 12.57%] | 13.94% | [12.60%, 15.32%] |
| Additives | D7 | 1 of 10,206 | 0.01% | [0.00%, 0.03%] | 0.01% | [0.00%, 0.02%] |
| Additives | at least one defect | 1,219 of 10,206 | 11.94% | [11.22%, 12.69%] | 14.07% | [12.74%, 15.46%] |
| CAPEX & SOCS | D1 class (absent receipt) | 46 of 4,672 | 0.98% | [0.62%, 1.40%] | 0.74% | [0.38%, 1.25%] |
| CAPEX & SOCS | D2 class (stale receipt) | 9 of 4,672 | 0.19% | [0.00%, 0.50%] | 0.10% | [0.00%, 0.31%] |
| CAPEX & SOCS | D3 | 15 of 4,672 | 0.32% | [0.11%, 0.59%] | 0.06% | [0.01%, 0.13%] |
| CAPEX & SOCS | D4 | 278 of 4,672 | 5.95% | [4.50%, 7.90%] | 25.55% | [12.52%, 38.45%] |
| CAPEX & SOCS | at least one defect | 332 of 4,672 | 7.11% | [5.57%, 9.12%] | 26.35% | [13.48%, 39.09%] |
| Chemicals & Intermediates | at least one defect | 0 of 1 | 0.00% | [0.00%, 0.00%] | 0.00% | [0.00%, 0.00%] |
| Commodity Resins | D1 class (absent receipt) | 1 of 306 | 0.33% | [0.00%, 1.10%] | 0.79% | [0.00%, 2.69%] |
| Commodity Resins | D4 | 80 of 306 | 26.14% | [19.93%, 32.47%] | 27.89% | [19.61%, 36.60%] |
| Commodity Resins | at least one defect | 81 of 306 | 26.47% | [20.18%, 32.78%] | 28.67% | [20.42%, 37.47%] |
| Energy | at least one defect | 0 of 3 | 0.00% | [0.00%, 0.00%] | 0.00% | [0.00%, 0.00%] |
| Enterprise Services | D1 class (absent receipt) | 1 of 370 | 0.27% | [0.00%, 0.90%] | 0.09% | [0.00%, 0.39%] |
| Enterprise Services | D2 class (stale receipt) | 1 of 370 | 0.27% | [0.00%, 0.93%] | 0.01% | [0.00%, 0.03%] |
| Enterprise Services | D4 | 20 of 370 | 5.41% | [2.55%, 9.09%] | 7.21% | [1.89%, 18.58%] |
| Enterprise Services | at least one defect | 22 of 370 | 5.95% | [2.98%, 9.83%] | 7.30% | [1.95%, 18.80%] |
| Latex & Monomers | D1 class (absent receipt) | 7 of 2,584 | 0.27% | [0.07%, 0.54%] | 0.26% | [0.05%, 0.55%] |
| Latex & Monomers | D2 class (stale receipt) | 3 of 2,584 | 0.12% | [0.00%, 0.27%] | 0.15% | [0.00%, 0.36%] |
| Latex & Monomers | D3 | 1 of 2,584 | 0.04% | [0.00%, 0.12%] | 0.01% | [0.00%, 0.02%] |
| Latex & Monomers | D4 | 479 of 2,584 | 18.54% | [16.86%, 20.28%] | 18.46% | [16.67%, 20.40%] |
| Latex & Monomers | D7 | 1 of 2,584 | 0.04% | [0.00%, 0.12%] | 0.05% | [0.00%, 0.16%] |
| Latex & Monomers | at least one defect | 488 of 2,584 | 18.89% | [17.18%, 20.63%] | 18.82% | [17.02%, 20.76%] |
| Logistics | D3 | 1 of 2,069 | 0.05% | [0.00%, 0.16%] | 0.01% | [0.00%, 0.03%] |
| Logistics | D4 | 114 of 2,069 | 5.51% | [3.09%, 8.13%] | 6.11% | [1.88%, 15.39%] |
| Logistics | at least one defect | 114 of 2,069 | 5.51% | [3.09%, 8.13%] | 6.11% | [1.88%, 15.39%] |
| Marketing | D1 class (absent receipt) | 6 of 1,220 | 0.49% | [0.00%, 1.31%] | 0.03% | [0.00%, 0.12%] |
| Marketing | D4 | 108 of 1,220 | 8.85% | [5.94%, 12.34%] | 47.19% | [25.71%, 59.53%] |

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Marketing | at least one defect | 114 of 1,220 | 9.34% | [6.36%, 12.92%] | 47.22% | [25.76%, 59.54%] |
| Others | D1 class (absent receipt) | 8 of 1,776 | 0.45% | [0.00%, 1.24%] | 0.00% | [0.00%, 0.00%] |
| Others | D4 | 86 of 1,776 | 4.84% | [3.52%, 6.31%] | 9.16% | [2.55%, 22.87%] |
| Others | at least one defect | 94 of 1,776 | 5.29% | [3.85%, 6.89%] | 9.16% | [2.55%, 22.88%] |
| Packaging | D1 class (absent receipt) | 136 of 63,561 | 0.21% | [0.16%, 0.27%] | 0.24% | [0.12%, 0.39%] |
| Packaging | D2 class (stale receipt) | 34 of 63,561 | 0.05% | [0.00%, 0.15%] | 0.04% | [0.00%, 0.09%] |
| Packaging | D3 | 15 of 63,561 | 0.02% | [0.01%, 0.04%] | 0.02% | [0.01%, 0.05%] |
| Packaging | D4 | 6,901 of 63,561 | 10.86% | [10.31%, 11.45%] | 18.95% | [17.63%, 20.33%] |
| Packaging | D7 | 4 of 63,561 | 0.01% | [0.00%, 0.01%] | 0.00% | [0.00%, 0.01%] |
| Packaging | at least one defect | 7,035 of 63,561 | 11.07% | [10.52%, 11.66%] | 19.13% | [17.82%, 20.53%] |
| Pigments & Colorants | D1 class (absent receipt) | 3 of 1,694 | 0.18% | [0.00%, 0.41%] | 0.24% | [0.00%, 0.67%] |
| Pigments & Colorants | D2 class (stale receipt) | 2 of 1,694 | 0.12% | [0.00%, 0.37%] | 0.19% | [0.00%, 0.60%] |
| Pigments & Colorants | D3 | 1 of 1,694 | 0.06% | [0.00%, 0.19%] | 0.17% | [0.00%, 0.56%] |
| Pigments & Colorants | D4 | 295 of 1,694 | 17.41% | [15.35%, 19.59%] | 29.19% | [23.93%, 34.78%] |
| Pigments & Colorants | at least one defect | 297 of 1,694 | 17.53% | [15.48%, 19.71%] | 29.37% | [24.09%, 34.95%] |
| Real Estate | D4 | 5 of 151 | 3.31% | [0.60%, 7.25%] | 12.38% | [0.45%, 25.30%] |
| Real Estate | at least one defect | 5 of 151 | 3.31% | [0.60%, 7.25%] | 12.38% | [0.45%, 25.30%] |
| Sales | D1 class (absent receipt) | 126 of 46,891 | 0.27% | [0.17%, 0.41%] | 0.32% | [0.12%, 0.59%] |
| Sales | D2 class (stale receipt) | 14 of 46,891 | 0.03% | [0.00%, 0.08%] | 0.02% | [0.00%, 0.05%] |
| Sales | D3 | 62 of 46,891 | 0.13% | [0.05%, 0.26%] | 0.10% | [0.04%, 0.19%] |
| Sales | D4 | 3,244 of 46,891 | 6.92% | [6.34%, 7.55%] | 15.91% | [13.78%, 18.18%] |
| Sales | D7 | 10 of 46,891 | 0.02% | [0.00%, 0.05%] | 0.05% | [0.00%, 0.14%] |
| Sales | at least one defect | 3,378 of 46,891 | 7.20% | [6.61%, 7.84%] | 16.24% | [14.14%, 18.48%] |
| Solvents | D3 | 3 of 1,573 | 0.19% | [0.00%, 0.43%] | 0.04% | [0.00%, 0.09%] |
| Solvents | D4 | 216 of 1,573 | 13.73% | [11.80%, 15.69%] | 16.61% | [13.63%, 19.78%] |
| Solvents | at least one defect | 219 of 1,573 | 13.92% | [12.00%, 15.87%] | 16.64% | [13.67%, 19.81%] |
| Specialty Resins | D4 | 228 of 1,295 | 17.61% | [15.33%, 19.94%] | 17.00% | [14.21%, 19.98%] |
| Specialty Resins | at least one defect | 228 of 1,295 | 17.61% | [15.33%, 19.94%] | 17.00% | [14.21%, 19.98%] |
| Spend Area Unidentified | D4 | 3 of 18 | 16.67% | [0.00%, 40.00%] | 9.65% | [0.00%, 30.06%] |
| Spend Area Unidentified | at least one defect | 3 of 18 | 16.67% | [0.00%, 40.00%] | 9.65% | [0.00%, 30.06%] |
| Titanium Dioxides | D1 class (absent receipt) | 3 of 429 | 0.70% | [0.00%, 1.59%] | 0.10% | [0.00%, 0.27%] |
| Titanium Dioxides | D2 class (stale receipt) | 2 of 429 | 0.47% | [0.00%, 1.21%] | 1.66% | [0.00%, 4.28%] |

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Titanium Dioxides | D4 | 119 of 429 | 27.74% | [22.84%, 32.89%] | 36.36% | [28.19%, 44.61%] |
| Titanium Dioxides | D6 | 2 of 429 | 0.47% | [0.00%, 1.51%] | 0.59% | [0.00%, 1.98%] |
| Titanium Dioxides | at least one defect | 124 of 429 | 28.90% | [23.95%, 34.11%] | 37.05% | [28.83%, 45.32%] |
| Trading & End Products | D1 class (absent receipt) | 261 of 15,104 | 1.73% | [0.93%, 2.68%] | 0.07% | [0.03%, 0.13%] |
| Trading & End Products | D2 class (stale receipt) | 12 of 15,104 | 0.08% | [0.01%, 0.16%] | 0.13% | [0.02%, 0.28%] |
| Trading & End Products | D3 | 259 of 15,104 | 1.71% | [0.93%, 2.68%] | 0.17% | [0.05%, 0.32%] |
| Trading & End Products | D4 | 3,339 of 15,104 | 22.11% | [18.90%, 25.57%] | 43.14% | [38.19%, 47.92%] |
| Trading & End Products | D7 | 3 of 15,104 | 0.02% | [0.00%, 0.05%] | 0.01% | [0.00%, 0.04%] |
| Trading & End Products | at least one defect | 3,556 of 15,104 | 23.54% | [20.24%, 27.05%] | 43.47% | [38.53%, 48.25%] |
| Workforce Services | D4 | 62 of 281 | 22.06% | [15.93%, 28.45%] | 28.17% | [19.43%, 36.45%] |
| Workforce Services | at least one defect | 62 of 281 | 22.06% | [15.93%, 28.45%] | 28.17% | [19.43%, 36.45%] |
| missing | D1 class (absent receipt) | 3 of 1,330 | 0.23% | [0.00%, 0.51%] | 0.17% | [0.00%, 0.44%] |
| missing | D2 class (stale receipt) | 1 of 1,330 | 0.08% | [0.00%, 0.24%] | 0.33% | [0.00%, 1.04%] |
| missing | D3 | 1 of 1,330 | 0.08% | [0.00%, 0.24%] | 0.01% | [0.00%, 0.03%] |
| missing | D4 | 373 of 1,330 | 28.05% | [25.10%, 30.94%] | 28.16% | [23.41%, 32.95%] |
| missing | D6 | 2 of 1,330 | 0.15% | [0.00%, 0.47%] | 0.04% | [0.00%, 0.12%] |
| missing | at least one defect | 377 of 1,330 | 28.35% | [25.34%, 31.29%] | 28.54% | [23.77%, 33.37%] |

### Month of the clearing

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| 2018-01 | D1 class (absent receipt) | 1 of 255 | 0.39% | [0.00%, 1.68%] | 3.62% | [0.00%, 12.12%] |
| 2018-01 | D3 | 1 of 255 | 0.39% | [0.00%, 1.68%] | 3.62% | [0.00%, 12.12%] |
| 2018-01 | D4 | 8 of 255 | 3.14% | [0.73%, 7.95%] | 9.26% | [0.10%, 22.02%] |
| 2018-01 | at least one defect | 9 of 255 | 3.53% | [0.94%, 8.63%] | 12.88% | [0.34%, 26.75%] |
| 2018-02 | D1 class (absent receipt) | 5 of 2,812 | 0.18% | [0.00%, 0.41%] | 0.01% | [0.00%, 0.02%] |
| 2018-02 | D2 class (stale receipt) | 1 of 2,812 | 0.04% | [0.00%, 0.12%] | 0.34% | [0.00%, 1.31%] |
| 2018-02 | D3 | 2 of 2,812 | 0.07% | [0.00%, 0.19%] | 0.03% | [0.00%, 0.12%] |
| 2018-02 | D4 | 96 of 2,812 | 3.41% | [2.60%, 4.32%] | 2.37% | [1.21%, 4.41%] |
| 2018-02 | at least one defect | 103 of 2,812 | 3.66% | [2.83%, 4.59%] | 2.75% | [1.38%, 5.13%] |
| 2018-03 | D1 class (absent receipt) | 19 of 9,790 | 0.19% | [0.07%, 0.36%] | 0.06% | [0.01%, 0.12%] |

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| 2018-03 | D2 class (stale receipt) | 4 of 9,790 | 0.04% | [0.00%, 0.10%] | 0.10% | [0.00%, 0.29%] |
| 2018-03 | D3 | 11 of 9,790 | 0.11% | [0.02%, 0.26%] | 0.04% | [0.00%, 0.11%] |
| 2018-03 | D4 | 855 of 9,790 | 8.73% | [7.37%, 10.37%] | 8.36% | [6.02%, 11.75%] |
| 2018-03 | D7 | 2 of 9,790 | 0.02% | [0.00%, 0.05%] | 0.00% | [0.00%, 0.02%] |
| 2018-03 | at least one defect | 877 of 9,790 | 8.96% | [7.58%, 10.61%] | 8.51% | [6.14%, 11.95%] |
| 2018-04 | D1 class (absent receipt) | 87 of 15,063 | 0.58% | [0.21%, 1.06%] | 0.15% | [0.01%, 0.36%] |
| 2018-04 | D2 class (stale receipt) | 6 of 15,063 | 0.04% | [0.01%, 0.07%] | 0.02% | [0.00%, 0.07%] |
| 2018-04 | D3 | 73 of 15,063 | 0.48% | [0.13%, 0.97%] | 0.01% | [0.00%, 0.02%] |
| 2018-04 | D4 | 1,737 of 15,063 | 11.53% | [10.29%, 12.95%] | 15.08% | [12.24%, 18.36%] |
| 2018-04 | D6 | 2 of 15,063 | 0.01% | [0.00%, 0.04%] | 0.32% | [0.00%, 1.06%] |
| 2018-04 | D7 | 2 of 15,063 | 0.01% | [0.00%, 0.04%] | 0.01% | [0.00%, 0.03%] |
| 2018-04 | at least one defect | 1,827 of 15,063 | 12.13% | [10.83%, 13.61%] | 15.54% | [12.63%, 18.94%] |
| 2018-05 | D1 class (absent receipt) | 60 of 18,939 | 0.32% | [0.15%, 0.58%] | 0.09% | [0.03%, 0.19%] |
| 2018-05 | D2 class (stale receipt) | 8 of 18,939 | 0.04% | [0.01%, 0.10%] | 0.03% | [0.00%, 0.07%] |
| 2018-05 | D3 | 38 of 18,939 | 0.20% | [0.05%, 0.46%] | 0.03% | [0.01%, 0.06%] |
| 2018-05 | D4 | 1,909 of 18,939 | 10.08% | [9.27%, 10.95%] | 16.92% | [10.76%, 25.85%] |
| 2018-05 | D7 | 1 of 18,939 | 0.01% | [0.00%, 0.02%] | 0.00% | [0.00%, 0.01%] |
| 2018-05 | at least one defect | 1,972 of 18,939 | 10.41% | [9.57%, 11.30%] | 17.03% | [10.85%, 25.97%] |
| 2018-06 | D1 class (absent receipt) | 70 of 19,748 | 0.35% | [0.20%, 0.53%] | 0.10% | [0.03%, 0.21%] |
| 2018-06 | D2 class (stale receipt) | 9 of 19,748 | 0.05% | [0.01%, 0.09%] | 0.10% | [0.01%, 0.25%] |
| 2018-06 | D3 | 44 of 19,748 | 0.22% | [0.08%, 0.39%] | 0.01% | [0.00%, 0.01%] |
| 2018-06 | D4 | 2,261 of 19,748 | 11.45% | [10.30%, 12.69%] | 15.09% | [11.54%, 19.60%] |
| 2018-06 | at least one defect | 2,331 of 19,748 | 11.80% | [10.64%, 13.06%] | 15.28% | [11.68%, 19.82%] |
| 2018-07 | D1 class (absent receipt) | 74 of 20,019 | 0.37% | [0.20%, 0.61%] | 0.07% | [0.02%, 0.17%] |
| 2018-07 | D2 class (stale receipt) | 1 of 20,019 | 0.00% | [0.00%, 0.02%] | 0.00% | [0.00%, 0.00%] |
| 2018-07 | D3 | 32 of 20,019 | 0.16% | [0.03%, 0.37%] | 0.01% | [0.00%, 0.01%] |
| 2018-07 | D4 | 1,991 of 20,019 | 9.95% | [8.93%, 11.07%] | 17.30% | [13.42%, 21.79%] |
| 2018-07 | D6 | 2 of 20,019 | 0.01% | [0.00%, 0.03%] | 0.00% | [0.00%, 0.01%] |
| 2018-07 | D7 | 1 of 20,019 | 0.00% | [0.00%, 0.02%] | 0.00% | [0.00%, 0.01%] |
| 2018-07 | at least one defect | 2,060 of 20,019 | 10.29% | [9.26%, 11.43%] | 17.38% | [13.49%, 21.90%] |
| 2018-08 | D1 class (absent receipt) | 56 of 21,610 | 0.26% | [0.18%, 0.35%] | 0.13% | [0.06%, 0.23%] |
| 2018-08 | D2 class (stale receipt) | 40 of 21,610 | 0.19% | [0.02%, 0.48%] | 0.07% | [0.01%, 0.15%] |
| 2018-08 | D3 | 13 of 21,610 | 0.06% | [0.02%, 0.11%] | 0.03% | [0.00%, 0.08%] |
| 2018-08 | D4 | 2,442 of 21,610 | 11.30% | [10.18%, 12.58%] | 22.47% | [17.18%, 28.35%] |
| 2018-08 | at least one defect | 2,531 of 21,610 | 11.71% | [10.55%, 13.02%] | 22.66% | [17.38%, 28.56%] |

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| 2018-09 | D1 class (absent receipt) | 77 of 19,002 | 0.41% | [0.26%, 0.58%] | 0.33% | [0.15%, 0.56%] |
| 2018-09 | D2 class (stale receipt) | 1 of 19,002 | 0.01% | [0.00%, 0.02%] | 0.00% | [0.00%, 0.00%] |
| 2018-09 | D3 | 23 of 19,002 | 0.12% | [0.03%, 0.23%] | 0.06% | [0.01%, 0.14%] |
| 2018-09 | D4 | 2,253 of 19,002 | 11.86% | [10.83%, 12.97%] | 27.43% | [23.69%, 31.34%] |
| 2018-09 | D7 | 5 of 19,002 | 0.03% | [0.01%, 0.06%] | 0.05% | [0.00%, 0.16%] |
| 2018-09 | at least one defect | 2,314 of 19,002 | 12.18% | [11.14%, 13.29%] | 27.67% | [23.93%, 31.56%] |
| 2018-10 | D1 class (absent receipt) | 91 of 19,193 | 0.47% | [0.15%, 0.96%] | 0.06% | [0.02%, 0.11%] |
| 2018-10 | D2 class (stale receipt) | 9 of 19,193 | 0.05% | [0.00%, 0.15%] | 0.00% | [0.00%, 0.01%] |
| 2018-10 | D3 | 64 of 19,193 | 0.33% | [0.03%, 0.82%] | 0.03% | [0.01%, 0.07%] |
| 2018-10 | D4 | 2,387 of 19,193 | 12.44% | [10.93%, 14.41%] | 26.80% | [21.62%, 32.61%] |
| 2018-10 | D7 | 4 of 19,193 | 0.02% | [0.00%, 0.05%] | 0.01% | [0.00%, 0.04%] |
| 2018-10 | at least one defect | 2,483 of 19,193 | 12.94% | [11.37%, 14.95%] | 26.87% | [21.69%, 32.69%] |
| 2018-11 | D1 class (absent receipt) | 70 of 9,103 | 0.77% | [0.15%, 1.70%] | 0.02% | [0.01%, 0.05%] |
| 2018-11 | D2 class (stale receipt) | 3 of 9,103 | 0.03% | [0.00%, 0.08%] | 1.29% | [0.00%, 3.45%] |
| 2018-11 | D3 | 61 of 9,103 | 0.67% | [0.05%, 1.60%] | 0.04% | [0.00%, 0.10%] |
| 2018-11 | D4 | 1,218 of 9,103 | 13.38% | [11.90%, 15.02%] | 22.30% | [15.43%, 30.79%] |
| 2018-11 | D7 | 4 of 9,103 | 0.04% | [0.00%, 0.10%] | 0.01% | [0.00%, 0.05%] |
| 2018-11 | at least one defect | 1,241 of 9,103 | 13.63% | [12.14%, 15.29%] | 22.36% | [15.48%, 30.86%] |

### Line value quartile

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Q1 | D1 class (absent receipt) | 369 of 38,549 | 0.96% | [0.63%, 1.34%] | 0.59% | [0.42%, 0.79%] |
| Q1 | D2 class (stale receipt) | 17 of 38,549 | 0.04% | [0.01%, 0.09%] | 0.05% | [0.01%, 0.12%] |
| Q1 | D3 | 273 of 38,549 | 0.71% | [0.40%, 1.08%] | 0.32% | [0.18%, 0.51%] |
| Q1 | D4 | 2,012 of 38,549 | 5.22% | [4.42%, 6.30%] | 5.01% | [4.45%, 5.64%] |
| Q1 | D7 | 3 of 38,549 | 0.01% | [0.00%, 0.02%] | 0.01% | [0.00%, 0.03%] |
| Q1 | at least one defect | 2,334 of 38,549 | 6.05% | [5.20%, 7.15%] | 5.60% | [5.01%, 6.26%] |
| Q2 | D1 class (absent receipt) | 91 of 38,481 | 0.24% | [0.17%, 0.32%] | 0.23% | [0.16%, 0.31%] |
| Q2 | D2 class (stale receipt) | 18 of 38,481 | 0.05% | [0.01%, 0.11%] | 0.04% | [0.01%, 0.09%] |
| Q2 | D3 | 35 of 38,481 | 0.09% | [0.04%, 0.16%] | 0.09% | [0.04%, 0.16%] |
| Q2 | D4 | 2,629 of 38,481 | 6.83% | [6.40%, 7.28%] | 7.01% | [6.57%, 7.46%] |
| Q2 | D7 | 3 of 38,481 | 0.01% | [0.00%, 0.02%] | 0.01% | [0.00%, 0.02%] |
| Q2 | at least one defect | 2,724 of 38,481 | 7.08% | [6.64%, 7.54%] | 7.24% | [6.79%, 7.71%] |
| Q3 | D1 class (absent receipt) | 67 of 38,963 | 0.17% | [0.12%, 0.23%] | 0.17% | [0.12%, 0.23%] |

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Q3 | D2 class (stale receipt) | 20 of 38,963 | 0.05% | [0.01%, 0.11%] | 0.04% | [0.01%, 0.08%] |
| Q3 | D3 | 29 of 38,963 | 0.07% | [0.04%, 0.12%] | 0.07% | [0.04%, 0.12%] |
| Q3 | D4 | 4,447 of 38,963 | 11.41% | [10.77%, 12.11%] | 12.35% | [11.60%, 13.17%] |
| Q3 | D7 | 3 of 38,963 | 0.01% | [0.00%, 0.02%] | 0.01% | [0.00%, 0.02%] |
| Q3 | at least one defect | 4,522 of 38,963 | 11.61% | [10.96%, 12.31%] | 12.53% | [11.78%, 13.34%] |
| Q4 | D1 class (absent receipt) | 83 of 39,541 | 0.21% | [0.15%, 0.27%] | 0.11% | [0.07%, 0.16%] |
| Q4 | D2 class (stale receipt) | 27 of 39,541 | 0.07% | [0.04%, 0.10%] | 0.13% | [0.03%, 0.29%] |
| Q4 | D3 | 25 of 39,541 | 0.06% | [0.03%, 0.10%] | 0.02% | [0.01%, 0.04%] |
| Q4 | D4 | 8,069 of 39,541 | 20.41% | [19.34%, 21.54%] | 19.90% | [16.24%, 23.97%] |
| Q4 | D6 | 4 of 39,541 | 0.01% | [0.00%, 0.03%] | 0.03% | [0.00%, 0.11%] |
| Q4 | D7 | 10 of 39,541 | 0.03% | [0.01%, 0.05%] | 0.01% | [0.00%, 0.03%] |
| Q4 | at least one defect | 8,168 of 39,541 | 20.66% | [19.60%, 21.78%] | 20.07% | [16.39%, 24.17%] |

### Vendor activity quartile

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Q1 | D1 class (absent receipt) | 114 of 38,820 | 0.29% | [0.22%, 0.38%] | 0.09% | [0.05%, 0.14%] |
| Q1 | D2 class (stale receipt) | 26 of 38,820 | 0.07% | [0.03%, 0.11%] | 0.14% | [0.01%, 0.34%] |
| Q1 | D3 | 45 of 38,820 | 0.12% | [0.06%, 0.18%] | 0.03% | [0.01%, 0.05%] |
| Q1 | D4 | 6,056 of 38,820 | 15.60% | [14.97%, 16.25%] | 18.93% | [14.45%, 24.29%] |
| Q1 | D6 | 4 of 38,820 | 0.01% | [0.00%, 0.03%] | 0.04% | [0.00%, 0.15%] |
| Q1 | D7 | 10 of 38,820 | 0.03% | [0.01%, 0.06%] | 0.00% | [0.00%, 0.01%] |
| Q1 | at least one defect | 6,194 of 38,820 | 15.96% | [15.31%, 16.62%] | 19.08% | [14.58%, 24.46%] |
| Q2 | D1 class (absent receipt) | 99 of 39,580 | 0.25% | [0.14%, 0.41%] | 0.21% | [0.08%, 0.37%] |
| Q2 | D2 class (stale receipt) | 15 of 39,580 | 0.04% | [0.01%, 0.07%] | 0.12% | [0.04%, 0.23%] |
| Q2 | D3 | 45 of 39,580 | 0.11% | [0.03%, 0.25%] | 0.02% | [0.01%, 0.03%] |
| Q2 | D4 | 5,287 of 39,580 | 13.36% | [12.46%, 14.29%] | 21.45% | [19.39%, 23.62%] |
| Q2 | D7 | 5 of 39,580 | 0.01% | [0.00%, 0.03%] | 0.03% | [0.00%, 0.10%] |
| Q2 | at least one defect | 5,391 of 39,580 | 13.62% | [12.71%, 14.56%] | 21.74% | [19.69%, 23.90%] |
| Q3 | D1 class (absent receipt) | 317 of 42,763 | 0.74% | [0.45%, 1.09%] | 0.24% | [0.07%, 0.56%] |
| Q3 | D2 class (stale receipt) | 13 of 42,763 | 0.03% | [0.00%, 0.08%] | 0.01% | [0.00%, 0.03%] |
| Q3 | D3 | 269 of 42,763 | 0.63% | [0.35%, 0.97%] | 0.08% | [0.03%, 0.15%] |
| Q3 | D4 | 3,571 of 42,763 | 8.35% | [7.10%, 9.78%] | 28.36% | [23.47%, 33.40%] |
| Q3 | D7 | 2 of 42,763 | 0.00% | [0.00%, 0.01%] | 0.00% | [0.00%, 0.00%] |
| Q3 | at least one defect | 3,829 of 42,763 | 8.95% | [7.69%, 10.39%] | 28.48% | [23.60%, 33.49%] |

| stratum | defect class (failed predicate) | clearings | share | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| Q4 | D1 class (absent receipt) | 80 of 34,371 | 0.23% | [0.17%, 0.30%] | 0.11% | [0.06%, 0.18%] |
| Q4 | D2 class (stale receipt) | 28 of 34,371 | 0.08% | [0.00%, 0.25%] | 0.05% | [0.00%, 0.14%] |
| Q4 | D3 | 3 of 34,371 | 0.01% | [0.00%, 0.02%] | 0.02% | [0.00%, 0.06%] |
| Q4 | D4 | 2,243 of 34,371 | 6.53% | [5.97%, 7.15%] | 7.50% | [6.37%, 8.70%] |
| Q4 | D7 | 2 of 34,371 | 0.01% | [0.00%, 0.01%] | 0.01% | [0.00%, 0.02%] |
| Q4 | at least one defect | 2,334 of 34,371 | 6.79% | [6.21%, 7.43%] | 7.65% | [6.51%, 8.86%] |

## Appendix E: Consequence and gate effect by stratum

The H1 and H2 comparisons (Section 5.2) and the gate verdict against the corrective (Section 5.3), at W = 60 days, per stratum. Each cell gives the O rate with the corrective events and the clearings of its group; "none" marks an empty group. These are descriptive: the strata are reported beside the estimates and never modelled in. Company and vendor strata are in the slot files only, as hashed labels.

**Item category: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)**

| stratum | H1: D1/D2 failed | H1: three-way, all pass | H1 difference | H2: D4 failed | H2: D4 pass | H2 difference |
|---|---|---|---|---|---|---|
| 2-way | none | none | n/a | 0.00% (0 of 4) | 2.72% (5 of 184) | -0.0272 |
| 3-way, invoice after receipt | none | 4.68% (377 of 8,048) | n/a | 6.81% (108 of 1,585) | 4.69% (378 of 8,055) | 0.0212 |
| 3-way, invoice before receipt | 5.64% (39 of 692) | 0.70% (909 of 129,554) | 0.0493 | 1.41% (219 of 15,568) | 0.73% (949 of 130,138) | 0.0068 |

**Item category: gate by outcome (clearings, W = 60 days)**

| stratum | analysed clearings | gate would have held | held, corrective within W | held, no corrective | allowed, corrective within W | allowed, no corrective |
|---|---|---|---|---|---|---|
| 2-way | 188 | 4 (2.13%) | 0 | 4 | 5 | 179 |
| 3-way, invoice after receipt | 9,640 | 1,592 (16.51%) | 109 | 1,483 | 377 | 7,671 |
| 3-way, invoice before receipt | 145,706 | 16,152 (11.09%) | 259 | 15,893 | 909 | 128,645 |

**Spend area: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)**

| stratum | H1: D1/D2 failed | H1: three-way, all pass | H1 difference | H2: D4 failed | H2: D4 pass | H2 difference |
|---|---|---|---|---|---|---|
| Additives | 7.69% (1 of 13) | 1.02% (92 of 8,987) | 0.0667 | 1.82% (22 of 1,207) | 1.03% (93 of 8,999) | 0.0079 |
| CAPEX & SOCS | 1.82% (1 of 55) | 1.45% (63 of 4,331) | 0.0036 | 2.16% (6 of 278) | 1.46% (64 of 4,394) | 0.0070 |
| Chemicals & Intermediates | none | 0.00% (0 of 1) | n/a | none | 0.00% (0 of 1) | n/a |

| stratum | H1: D1/D2 failed | H1: three-way, all pass | H1 difference | H2: D4 failed | H2: D4 pass | H2 difference |
|---|---|---|---|---|---|---|
| Commodity Resins | 0.00% (0 of 1) | 2.22% (5 of 225) | -0.0222 | 1.25% (1 of 80) | 2.21% (5 of 226) | -0.0096 |
| Energy | none | none | n/a | none | 0.00% (0 of 3) | n/a |
| Enterprise Services | 0.00% (0 of 2) | 2.05% (7 of 342) | -0.0205 | 0.00% (0 of 20) | 2.00% (7 of 350) | -0.0200 |
| Latex & Monomers | 30.00% (3 of 10) | 1.96% (41 of 2,096) | 0.2804 | 1.88% (9 of 479) | 2.09% (44 of 2,105) | -0.0021 |
| Logistics | none | 10.49% (205 of 1,955) | n/a | 35.96% (41 of 114) | 10.49% (205 of 1,955) | 0.2548 |
| Marketing | 16.67% (1 of 6) | 2.98% (33 of 1,106) | 0.1368 | 12.04% (13 of 108) | 3.06% (34 of 1,112) | 0.0898 |
| Others | 0.00% (0 of 8) | 1.44% (24 of 1,663) | -0.0144 | 5.81% (5 of 86) | 1.54% (26 of 1,690) | 0.0428 |
| Packaging | 5.29% (9 of 170) | 0.45% (254 of 56,526) | 0.0484 | 1.10% (76 of 6,901) | 0.46% (263 of 56,660) | 0.0064 |
| Pigments & Colorants | 40.00% (2 of 5) | 1.22% (17 of 1,397) | 0.3878 | 2.71% (8 of 295) | 1.29% (18 of 1,399) | 0.0143 |
| Real Estate | none | 0.00% (0 of 1) | n/a | 0.00% (0 of 5) | 2.05% (3 of 146) | -0.0205 |
| Sales | 13.57% (19 of 140) | 0.73% (318 of 43,511) | 0.1284 | 1.02% (33 of 3,244) | 0.78% (340 of 43,647) | 0.0024 |
| Solvents | none | 1.55% (21 of 1,354) | n/a | 3.24% (7 of 216) | 1.55% (21 of 1,357) | 0.0169 |
| Specialty Resins | none | 1.03% (11 of 1,067) | n/a | 3.51% (8 of 228) | 1.03% (11 of 1,067) | 0.0248 |
| Spend Area Unidentified | none | 0.00% (0 of 15) | n/a | 0.00% (0 of 3) | 0.00% (0 of 15) | 0.0000 |
| Titanium Dioxides | 0.00% (0 of 5) | 2.62% (8 of 305) | -0.0262 | 7.56% (9 of 119) | 2.58% (8 of 310) | 0.0498 |
| Trading & End Products | 0.73% (2 of 273) | 1.22% (141 of 11,548) | -0.0049 | 2.13% (71 of 3,339) | 1.22% (143 of 11,765) | 0.0091 |
| Workforce Services | none | 16.89% (37 of 219) | n/a | 17.74% (11 of 62) | 16.89% (37 of 219) | 0.0085 |
| missing | 25.00% (1 of 4) | 0.94% (9 of 953) | 0.2406 | 1.88% (7 of 373) | 1.04% (10 of 957) | 0.0083 |

**Spend area: gate by outcome (clearings, W = 60 days)**

| stratum | analysed clearings | gate would have held | held, corrective within W | held, no corrective | allowed, corrective within W | allowed, no corrective |
|---|---|---|---|---|---|---|
| Additives | 10,206 | 1,219 (11.94%) | 23 | 1,196 | 92 | 8,895 |
| CAPEX & SOCS | 4,672 | 332 (7.11%) | 7 | 325 | 63 | 4,277 |
| Chemicals & Intermediates | 1 | 0 (0.00%) | 0 | 0 | 0 | 1 |
| Commodity Resins | 306 | 81 (26.47%) | 1 | 80 | 5 | 220 |
| Energy | 3 | 0 (0.00%) | 0 | 0 | 0 | 3 |
| Enterprise Services | 370 | 22 (5.95%) | 0 | 22 | 7 | 341 |
| Latex & Monomers | 2,584 | 488 (18.89%) | 12 | 476 | 41 | 2,055 |
| Logistics | 2,069 | 114 (5.51%) | 41 | 73 | 205 | 1,750 |
| Marketing | 1,220 | 114 (9.34%) | 14 | 100 | 33 | 1,073 |
| Others | 1,776 | 94 (5.29%) | 5 | 89 | 26 | 1,656 |
| Packaging | 63,561 | 7,035 (11.07%) | 85 | 6,950 | 254 | 56,272 |
| Pigments & Colorants | 1,694 | 297 (17.53%) | 9 | 288 | 17 | 1,380 |
| Real Estate | 151 | 5 (3.31%) | 0 | 5 | 3 | 143 |
| Sales | 46,891 | 3,378 (7.20%) | 55 | 3,323 | 318 | 43,195 |
| Solvents | 1,573 | 219 (13.92%) | 7 | 212 | 21 | 1,333 |

| stratum | analysed clearings | gate would have held | held, corrective within W | held, no corrective | allowed, corrective within W | allowed, no corrective |
|---|---|---|---|---|---|---|
| Specialty Resins | 1,295 | 228 (17.61%) | 8 | 220 | 11 | 1,056 |
| Spend Area Unidentified | 18 | 3 (16.67%) | 0 | 3 | 0 | 15 |
| Titanium Dioxides | 429 | 124 (28.90%) | 9 | 115 | 8 | 297 |
| Trading & End Products | 15,104 | 3,556 (23.54%) | 73 | 3,483 | 141 | 11,407 |
| Workforce Services | 281 | 62 (22.06%) | 11 | 51 | 37 | 182 |
| missing | 1,330 | 377 (28.35%) | 8 | 369 | 9 | 944 |

**Month of the clearing: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)**

| stratum | H1: D1/D2 failed | H1: three-way, all pass | H1 difference | H2: D4 failed | H2: D4 pass | H2 difference |
|---|---|---|---|---|---|---|
| 2018-01 | 0.00% (0 of 1) | 0.00% (0 of 238) | 0.0000 | 0.00% (0 of 8) | 0.00% (0 of 247) | 0.0000 |
| 2018-02 | 0.00% (0 of 6) | 0.30% (8 of 2,703) | -0.0030 | 2.08% (2 of 96) | 0.29% (8 of 2,716) | 0.0179 |
| 2018-03 | 13.04% (3 of 23) | 0.88% (78 of 8,902) | 0.1217 | 2.92% (25 of 855) | 0.91% (81 of 8,935) | 0.0202 |
| 2018-04 | 3.23% (3 of 93) | 0.57% (76 of 13,223) | 0.0265 | 0.92% (16 of 1,737) | 0.59% (78 of 13,326) | 0.0034 |
| 2018-05 | 4.41% (3 of 68) | 0.82% (139 of 16,953) | 0.0359 | 1.36% (26 of 1,909) | 0.85% (144 of 17,030) | 0.0052 |
| 2018-06 | 2.53% (2 of 79) | 1.13% (197 of 17,400) | 0.0140 | 2.12% (48 of 2,261) | 1.14% (199 of 17,487) | 0.0098 |
| 2018-07 | 12.00% (9 of 75) | 1.20% (216 of 17,936) | 0.1080 | 1.81% (36 of 1,991) | 1.25% (225 of 18,028) | 0.0056 |
| 2018-08 | 6.25% (6 of 96) | 0.98% (186 of 19,054) | 0.0527 | 1.80% (44 of 2,442) | 1.00% (191 of 19,168) | 0.0081 |
| 2018-09 | 2.56% (2 of 78) | 0.87% (145 of 16,661) | 0.0169 | 2.44% (55 of 2,253) | 0.90% (151 of 16,749) | 0.0154 |
| 2018-10 | 9.00% (9 of 100) | 0.92% (153 of 16,683) | 0.0808 | 2.01% (48 of 2,387) | 0.98% (164 of 16,806) | 0.0104 |
| 2018-11 | 2.74% (2 of 73) | 1.12% (88 of 7,849) | 0.0162 | 2.22% (27 of 1,218) | 1.15% (91 of 7,885) | 0.0106 |

**Month of the clearing: gate by outcome (clearings, W = 60 days)**

| stratum | analysed clearings | gate would have held | held, corrective within W | held, no corrective | allowed, corrective within W | allowed, no corrective |
|---|---|---|---|---|---|---|
| 2018-01 | 255 | 9 (3.53%) | 0 | 9 | 0 | 246 |
| 2018-02 | 2,812 | 103 (3.66%) | 2 | 101 | 8 | 2,701 |
| 2018-03 | 9,790 | 877 (8.96%) | 28 | 849 | 78 | 8,835 |
| 2018-04 | 15,063 | 1,827 (12.13%) | 18 | 1,809 | 76 | 13,160 |
| 2018-05 | 18,939 | 1,972 (10.41%) | 29 | 1,943 | 141 | 16,826 |
| 2018-06 | 19,748 | 2,331 (11.80%) | 50 | 2,281 | 197 | 17,220 |
| 2018-07 | 20,019 | 2,060 (10.29%) | 45 | 2,015 | 216 | 17,743 |
| 2018-08 | 21,610 | 2,531 (11.71%) | 49 | 2,482 | 186 | 18,893 |
| 2018-09 | 19,002 | 2,314 (12.18%) | 58 | 2,256 | 148 | 16,540 |
| 2018-10 | 19,193 | 2,483 (12.94%) | 59 | 2,424 | 153 | 16,557 |
| 2018-11 | 9,103 | 1,241 (13.63%) | 30 | 1,211 | 88 | 7,774 |

**Line value quartile: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)**

| stratum | H1: D1/D2 failed | H1: three-way, all pass | H1 difference | H2: D4 failed | H2: D4 pass | H2 difference |
|---|---|---|---|---|---|---|
| Q1 | 2.59% (10 of 386) | 0.56% (203 of 36,210) | 0.0203 | 0.55% (11 of 2,012) | 0.58% (212 of 36,537) | -0.0003 |
| Q2 | 8.26% (9 of 109) | 0.59% (210 of 35,733) | 0.0767 | 0.61% (16 of 2,629) | 0.61% (219 of 35,852) | -0.0000 |
| Q3 | 5.75% (5 of 87) | 0.86% (296 of 34,382) | 0.0489 | 1.10% (49 of 4,447) | 0.87% (301 of 34,516) | 0.0023 |
| Q4 | 13.64% (15 of 110) | 1.84% (577 of 31,277) | 0.1179 | 3.11% (251 of 8,069) | 1.91% (600 of 31,472) | 0.0120 |

**Line value quartile: gate by outcome (clearings, W = 60 days)**

| stratum | analysed clearings | gate would have held | held, corrective within W | held, no corrective | allowed, corrective within W | allowed, no corrective |
|---|---|---|---|---|---|---|
| Q1 | 38,549 | 2,334 (6.05%) | 20 | 2,314 | 203 | 36,012 |
| Q2 | 38,481 | 2,724 (7.08%) | 25 | 2,699 | 210 | 35,547 |
| Q3 | 38,963 | 4,522 (11.61%) | 54 | 4,468 | 296 | 34,145 |
| Q4 | 39,541 | 8,168 (20.66%) | 269 | 7,899 | 582 | 30,791 |

**Vendor activity quartile: H1 and H2 (O rate, corrective events of clearings, at W = 60 days)**

| stratum | H1: D1/D2 failed | H1: three-way, all pass | H1 difference | H2: D4 failed | H2: D4 pass | H2 difference |
|---|---|---|---|---|---|---|
| Q1 | 11.43% (16 of 140) | 2.08% (676 of 32,442) | 0.0934 | 2.91% (176 of 6,056) | 2.13% (699 of 32,764) | 0.0077 |
| Q2 | 14.91% (17 of 114) | 0.76% (259 of 34,189) | 0.1415 | 1.91% (101 of 5,287) | 0.80% (276 of 34,293) | 0.0111 |
| Q3 | 1.21% (4 of 330) | 0.72% (280 of 38,934) | 0.0049 | 0.98% (35 of 3,571) | 0.72% (284 of 39,192) | 0.0026 |
| Q4 | 1.85% (2 of 108) | 0.22% (71 of 32,037) | 0.0163 | 0.67% (15 of 2,243) | 0.23% (73 of 32,128) | 0.0044 |

**Vendor activity quartile: gate by outcome (clearings, W = 60 days)**

| stratum | analysed clearings | gate would have held | held, corrective within W | held, no corrective | allowed, corrective within W | allowed, no corrective |
|---|---|---|---|---|---|---|
| Q1 | 38,820 | 6,194 (15.96%) | 194 | 6,000 | 681 | 31,945 |
| Q2 | 39,580 | 5,391 (13.62%) | 118 | 5,273 | 259 | 33,930 |
| Q3 | 42,763 | 3,829 (8.95%) | 39 | 3,790 | 280 | 38,654 |
| Q4 | 34,371 | 2,334 (6.79%) | 17 | 2,317 | 71 | 31,966 |

Sources: `results/e2/confounders.json`, `results/e3/confounders.json`.

## Appendix F: Model inputs (exploratory)

Exploratory and illustrative. A synthetic line from the test fixtures, never a real line, rendered by the code the agent replay used. The design does not separate wording, structure and information content: the two views differ in all three at once.

The registered template, `prereg/erp-view-template.txt`, verbatim (SHA-256 `11726f914f3d0ea79ae374f0bd2ba71c6e75ea60ec88ae13745b717ca563b48a`):

```
# prereg/erp-view-template.txt: registered serialisation of the ERP view for E4.
# Everything above the line "=== REQUEST ===" is documentation and is never sent.
# The request is the block between "=== REQUEST ===" and "=== END ===", sent
# verbatim as the single user message after substituting each {placeholder}.
#
# The view is the state of the line at the decision point t* (prereg/p7-v1.md
# section 7), built only from events at or before t*. It carries no event
# timeline, no company, no vendor, no actor name and no line id.
#
# Placeholders:
#   {category}            one of: 3-way match, invoice after goods receipt |
#                         3-way match, invoice before goods receipt | 2-way match
#   {gr_based_iv}         yes | no
#   {gr_required}         yes | no
#   {approved_price}      value on the last approval-bearing event (fallback: line value),
#                         anonymised EUR, 2 decimals
#   {approved_quantity}   the quantity field if the column mapping names one; the registered
#                         mapping names none, so the literal "not recorded in this system"
#   {invoice_received}    yes | no
#   {invoice_date}        YYYY-MM-DD of the latest Record Invoice Receipt, or "none"
#   {invoice_value}       value on that event, anonymised EUR, 2 decimals, or "none"
#   {gr_recorded}         yes | no   (uncancelled goods receipt count > 0)
#   {gr_date}             YYYY-MM-DD of the latest uncancelled Record Goods Receipt, or "none"
#   {changes}             "none", or a comma-separated list in time order of
#                         "price changed on YYYY-MM-DD" / "quantity changed on YYYY-MM-DD"
#                         for changes after the last approval-bearing event
#   {payment_block}       set | not set
#   {decision_date}       YYYY-MM-DD of t*
#
# The model's reply is parsed as paper 6 jev-v3 section 3b, with the prefill "{".
# Fields: decision in {"clear", "hold"}; p_hold a number in [0, 1].
=== REQUEST ===
You are the accounts-payable control for one purchase-order line. The invoice on this line is ready to be
↪  cleared for payment today. Using only the ERP record below, decide whether to clear the invoice for
↪  payment now or to hold it. Reply with only a JSON object of the form {"decision": "clear" or "hold",
↪  "p_hold": <number between 0 and 1>}, where p_hold is your probability that the invoice should be held. No
↪  code fences. No text after the object.

ERP record as of {decision_date}
Item category: {category}
GR-based invoice verification: {gr_based_iv}
Goods receipt required: {gr_required}
Approved price (EUR): {approved_price}
Approved quantity: {approved_quantity}
Invoice received: {invoice_received}
Invoice date: {invoice_date}
Invoice value (EUR): {invoice_value}
Goods receipt recorded: {gr_recorded}
Goods receipt date: {gr_date}
Changes since approval: {changes}
Payment block: {payment_block}
=== END ===
```

The synthetic line `B_stale_receipt_3way_before` (3-way, invoice before receipt) at its decision point 2018-03-10T09:00:00+00:00, gate verdict hold, failed predicates D1, D2.

The ERP view (registered):

```
You are the accounts-payable control for one purchase-order line. The invoice on this line is ready to be
↪  cleared for payment today. Using only the ERP record below, decide whether to clear the invoice for
↪  payment now or to hold it. Reply with only a JSON object of the form {"decision": "clear" or "hold",
↪  "p_hold": <number between 0 and 1>}, where p_hold is your probability that the invoice should be held. No
↪  code fences. No text after the object.

ERP record as of 2018-03-10
Item category: 3-way match, invoice before goods receipt
GR-based invoice verification: no
Goods receipt required: yes
```

```
Approved price (EUR): 200.00
Approved quantity: not recorded in this system
Invoice received: yes
Invoice date: 2018-03-05
Invoice value (EUR): 200.00
Goods receipt recorded: no
Goods receipt date: none
Changes since approval: none
Payment block: not set
```

The timeline view (exploratory), as corrected in 1.1.1: the events up to the decision point without the decision's own clearing; the opening paragraph is the template's, unchanged:

```
You are the accounts-payable control for one purchase-order line. The invoice on this line is ready to be
↪  cleared for payment today. Using only the ERP record below, decide whether to clear the invoice for
↪  payment now or to hold it. Reply with only a JSON object of the form {"decision": "clear" or "hold",
↪  "p_hold": <number between 0 and 1>}, where p_hold is your probability that the invoice should be held. No
↪  code fences. No text after the object.

Event timeline up to 2018-03-10 (date, activity, value in EUR)
2018-03-01, Create Purchase Order Item, 200.00
2018-03-05, Record Invoice Receipt, 200.00
```

The Haiku request body:


```
{
  "model": "claude-haiku-4-5-20251001",
  "max_tokens": 48,
  "temperature": 0.0,
  "messages": [
    {
      "role": "user",
      "content": "You are the accounts-payable control for one purchase-order line. The invoice on this line is
      ↪  ready to be cleared for payment today. Using only the ERP record below, decide whether to clear the
      ↪  invoice for payment now or to hold it. Reply with only a JSON object of the form {\"decision\":
      ↪  \"clear\" or \"hold\", \"p_hold\": <number between 0 and 1>}, where p_hold is your probability that
      ↪  the invoice should be held. No code fences. No text after the object.\n\nERP record as of
      ↪  2018-03-10\nItem category: 3-way match, invoice before goods receipt\nGR-based invoice verification:
      ↪  no\nGoods receipt required: yes\nApproved price (EUR): 200.00\nApproved quantity: not recorded in
      ↪  this system\nInvoice received: yes\nInvoice date: 2018-03-05\nInvoice value (EUR): 200.00\nGoods
      ↪  receipt recorded: no\nGoods receipt date: none\nChanges since approval: none\nPayment block: not set"
    },
    {
      "role": "assistant",
      "content": "{"
    }
  ]
}
```


The Jev request (the template's closing reply-format sentence (a JSON decision and hold probability) is omitted from the Jev context and replaced by a typed choice with the labels clear and hold (prereg/DEVIATIONS.md, E4 Jev request format)):


```
{
  "state": {
    "context": "You are the accounts-payable control for one purchase-order line. The invoice on this line is
    ↪  ready to be cleared for payment today. Using only the ERP record below, decide whether to clear the
    ↪  invoice for payment now or to hold it.",
    "record": "ERP record as of 2018-03-10\nItem category: 3-way match, invoice before goods receipt\nGR-based
    ↪  invoice verification: no\nGoods receipt required: yes\nApproved price (EUR): 200.00\nApproved quantity:
    ↪  not recorded in this system\nInvoice received: yes\nInvoice date: 2018-03-05\nInvoice value (EUR):
    ↪  200.00\nGoods receipt recorded: no\nGoods receipt date: none\nChanges since approval: none\nPayment
    ↪  block: not set"
  },
  "model": "jev-latest",
  "questions": {
    "Q_clear_or_hold": {
      "type": "choice",
      "instructions": "Decide whether to clear the invoice for payment now or to hold it.",
```

```
      "criteria": {
        "clear": "Clear the invoice for payment now.",
        "hold": "Hold the invoice; do not pay it now."
      }
    }
  }
}
```

| view | exposes | omits |
|---|---|---|
| ERP view | approved_price, approved_quantity, category, changes, decision_date, gr_based_iv, gr_date, gr_recorded, gr_required, invoice_date, invoice_received, invoice_value, payment_block | the event timeline; actor names; line, document, company and vendor identifiers |
| timeline view | the template's task paragraph, event date, activity, value in EUR | the ERP record's derived fields (approved price, invoice and receipt status, changes since approval, payment block); actor names; line, document, company and vendor identifiers |

## Appendix G: Exploratory analyses

The full tables of the exploratory analyses of Sections 5.1 to 5.3 (decision D-G-1): X1 by item category and H2 under the strict reading of D4; X2 per stratum, with each stratum's comparison weight and whether it is kept; X3 by item category and line-value quartile at both windows. Not registered; no p-value; outside the Holm family.

### X1, 2-way

| reading (exploratory) | clearings | share | Wilson 95% | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| (a) registered: at least one failed predicate | 4 of 188 | 2.13% | [0.83%, 5.34%] | [0.00%, 4.94%] | 2.68% | [0.00%, 8.71%] |
| (b) D4 counts only changes after an explicit approval or release event | 4 of 188 | 2.13% | [0.83%, 5.34%] | [0.00%, 4.94%] | 2.68% | [0.00%, 8.71%] |
| (c) D4 failures with the creation as the only approval-bearing event | 0 of 188 | 0.00% | [0.00%, 2.00%] | [0.00%, 0.00%] | 0.00% | [0.00%, 0.00%] |

### X1, 3-way, invoice after receipt

| reading (exploratory) | clearings | share | Wilson 95% | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| (a) registered: at least one failed predicate | 1,592 of 9,640 | 16.51% | [15.79%, 17.27%] | [14.42%, 18.79%] | 15.67% | [9.50%, 24.66%] |

| reading (exploratory) | clearings | share | Wilson 95% | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| (b) D4 counts only changes after an explicit approval or release event | 133 of 9,640 | 1.38% | [1.17%, 1.63%] | [0.99%, 1.81%] | 9.48% | [4.85%, 16.39%] |
| (c) D4 failures with the creation as the only approval-bearing event | 1,461 of 9,640 | 15.16% | [14.45%, 15.89%] | [13.07%, 17.47%] | 6.20% | [3.28%, 11.00%] |

**X1, 3-way, invoice before receipt**

| reading (exploratory) | clearings | share | Wilson 95% | clustered 95% | share of value | clustered 95% |
|---|---|---|---|---|---|---|
| (a) registered: at least one failed predicate | 16,152 of 145,706 | 11.09% | [10.93%, 11.25%] | [10.60%, 11.60%] | 22.21% | [21.00%, 23.48%] |
| (b) D4 counts only changes after an explicit approval or release event | 1,268 of 145,706 | 0.87% | [0.82%, 0.92%] | [0.62%, 1.17%] | 2.86% | [1.89%, 3.99%] |
| (c) D4 failures with the creation as the only approval-bearing event | 15,017 of 145,706 | 10.31% | [10.15%, 10.46%] | [9.88%, 10.75%] | 19.41% | [18.47%, 20.36%] |

**X1, H2 under the explicit-anchor reading (W = 60 days)**

| | failing group | comparison group | difference | clustered 95% |
|---|---|---|---|---|
| exploratory (b) | 4.42% (30 of 679) | 1.05% (1,629 of 154,855) | 0.0337 | [0.0129, 0.0736] |
| registered | 1.91% | 0.96% | 0.0094 | |

**X2, H1 by category x line-value quartile** (standardised difference 0.0662, [0.0378, 0.0984])

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| 2-way, Q1 | n/a (0 of 0) | n/a (0 of 0) | n/a | 0.00% | no |
| 2-way, Q2 | n/a (0 of 0) | n/a (0 of 0) | n/a | 0.00% | no |
| 2-way, Q3 | n/a (0 of 0) | n/a (0 of 0) | n/a | 0.00% | no |
| 2-way, Q4 | n/a (0 of 0) | n/a (0 of 0) | n/a | 0.00% | no |
| 3-way, invoice after receipt, Q1 | n/a (0 of 0) | 0.78% (15 of 1,928) | n/a | 1.40% | no |
| 3-way, invoice after receipt, Q2 | n/a (0 of 0) | 1.39% (24 of 1,726) | n/a | 1.25% | no |
| 3-way, invoice after receipt, Q3 | n/a (0 of 0) | 3.41% (64 of 1,876) | n/a | 1.36% | no |
| 3-way, invoice after receipt, Q4 | n/a (0 of 0) | 10.88% (274 of 2,518) | n/a | 1.83% | no |
| 3-way, invoice before receipt, Q1 | 2.59% (10 of 386) | 0.55% (188 of 34,282) | 0.0204 | 24.91% | yes |

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| 3-way, invoice before receipt, Q2 | 8.26% (9 of 109) | 0.55% (186 of 34,007) | 0.0771 | 24.71% | yes |
| 3-way, invoice before receipt, Q3 | 5.75% (5 of 87) | 0.71% (232 of 32,506) | 0.0503 | 23.62% | yes |
| 3-way, invoice before receipt, Q4 | 13.64% (15 of 110) | 1.05% (303 of 28,759) | 0.1258 | 20.90% | yes |

**X2, H1 by month of t_c** (standardised difference 0.0507, [0.0290, 0.0824])

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| 2018-01 | 0.00% (0 of 1) | 0.00% (0 of 238) | 0.0000 | 0.17% | yes |
| 2018-02 | 0.00% (0 of 6) | 0.30% (8 of 2,703) | -0.0030 | 1.96% | yes |
| 2018-03 | 13.04% (3 of 23) | 0.88% (78 of 8,902) | 0.1217 | 6.47% | yes |
| 2018-04 | 3.23% (3 of 93) | 0.57% (76 of 13,223) | 0.0265 | 9.61% | yes |
| 2018-05 | 4.41% (3 of 68) | 0.82% (139 of 16,953) | 0.0359 | 12.32% | yes |
| 2018-06 | 2.53% (2 of 79) | 1.13% (197 of 17,400) | 0.0140 | 12.65% | yes |
| 2018-07 | 12.00% (9 of 75) | 1.20% (216 of 17,936) | 0.1080 | 13.03% | yes |
| 2018-08 | 6.25% (6 of 96) | 0.98% (186 of 19,054) | 0.0527 | 13.85% | yes |
| 2018-09 | 2.56% (2 of 78) | 0.87% (145 of 16,661) | 0.0169 | 12.11% | yes |
| 2018-10 | 9.00% (9 of 100) | 0.92% (153 of 16,683) | 0.0808 | 12.12% | yes |
| 2018-11 | 2.74% (2 of 73) | 1.12% (88 of 7,849) | 0.0162 | 5.70% | yes |

**X2, H1 by spend area** (standardised difference 0.0778, [0.0461, 0.1125])

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| Additives | 7.69% (1 of 13) | 1.02% (92 of 8,987) | 0.0667 | 6.53% | yes |
| CAPEX & SOCS | 1.82% (1 of 55) | 1.45% (63 of 4,331) | 0.0036 | 3.15% | yes |
| Chemicals & Intermediates | n/a (0 of 0) | 0.00% (0 of 1) | n/a | 0.00% | no |
| Commodity Resins | 0.00% (0 of 1) | 2.22% (5 of 225) | -0.0222 | 0.16% | yes |
| Energy | n/a (0 of 0) | n/a (0 of 0) | n/a | 0.00% | no |
| Enterprise Services | 0.00% (0 of 2) | 2.05% (7 of 342) | -0.0205 | 0.25% | yes |
| Latex & Monomers | 30.00% (3 of 10) | 1.96% (41 of 2,096) | 0.2804 | 1.52% | yes |
| Logistics | n/a (0 of 0) | 10.49% (205 of 1,955) | n/a | 1.42% | no |
| Marketing | 16.67% (1 of 6) | 2.98% (33 of 1,106) | 0.1368 | 0.80% | yes |
| Others | 0.00% (0 of 8) | 1.44% (24 of 1,663) | -0.0144 | 1.21% | yes |
| Packaging | 5.29% (9 of 170) | 0.45% (254 of 56,526) | 0.0484 | 41.08% | yes |
| Pigments & Colorants | 40.00% (2 of 5) | 1.22% (17 of 1,397) | 0.3878 | 1.02% | yes |
| Real Estate | n/a (0 of 0) | 0.00% (0 of 1) | n/a | 0.00% | no |
| Sales | 13.57% (19 of 140) | 0.73% (318 of 43,511) | 0.1284 | 31.62% | yes |
| Solvents | n/a (0 of 0) | 1.55% (21 of 1,354) | n/a | 0.98% | no |
| Specialty Resins | n/a (0 of 0) | 1.03% (11 of 1,067) | n/a | 0.78% | no |
| Spend Area Unidentified | n/a (0 of 0) | 0.00% (0 of 15) | n/a | 0.01% | no |
| Titanium Dioxides | 0.00% (0 of 5) | 2.62% (8 of 305) | -0.0262 | 0.22% | yes |
| Trading & End Products | 0.73% (2 of 273) | 1.22% (141 of 11,548) | -0.0049 | 8.39% | yes |
| Workforce Services | n/a (0 of 0) | 16.89% (37 of 219) | n/a | 0.16% | no |
| missing | 25.00% (1 of 4) | 0.94% (9 of 953) | 0.2406 | 0.69% | yes |

**X2, H2 by category x line-value quartile** (standardised difference 0.0031, [0.0008, 0.0054])

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| 2-way, Q1 | n/a (0 of 0) | 0.00% (0 of 5) | n/a | 0.00% | no |
| 2-way, Q2 | n/a (0 of 0) | 0.00% (0 of 24) | n/a | 0.02% | no |
| 2-way, Q3 | n/a (0 of 0) | 0.00% (0 of 59) | n/a | 0.04% | no |
| 2-way, Q4 | 0.00% (0 of 4) | 5.21% (5 of 96) | -0.0521 | 0.07% | yes |
| 3-way, invoice after receipt, Q1 | 0.29% (1 of 341) | 0.78% (15 of 1,930) | -0.0048 | 1.39% | yes |
| 3-way, invoice after receipt, Q2 | 0.36% (1 of 276) | 1.39% (24 of 1,726) | -0.0103 | 1.25% | yes |
| 3-way, invoice after receipt, Q3 | 1.60% (4 of 250) | 3.41% (64 of 1,876) | -0.0181 | 1.36% | yes |
| 3-way, invoice after receipt, Q4 | 14.21% (102 of 718) | 10.90% (275 of 2,523) | 0.0331 | 1.82% | yes |
| 3-way, invoice before receipt, Q1 | 0.60% (10 of 1,671) | 0.57% (197 of 34,602) | 0.0003 | 25.01% | yes |
| 3-way, invoice before receipt, Q2 | 0.64% (15 of 2,353) | 0.57% (195 of 34,102) | 0.0007 | 24.64% | yes |
| 3-way, invoice before receipt, Q3 | 1.07% (45 of 4,197) | 0.73% (237 of 32,581) | 0.0034 | 23.55% | yes |
| 3-way, invoice before receipt, Q4 | 2.03% (149 of 7,347) | 1.11% (320 of 28,853) | 0.0092 | 20.85% | yes |

**X2, H2 by month of t__c** (standardised difference 0.0094, [0.0060, 0.0132])

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| 2018-01 | 0.00% (0 of 8) | 0.00% (0 of 247) | 0.0000 | 0.18% | yes |
| 2018-02 | 2.08% (2 of 96) | 0.29% (8 of 2,716) | 0.0179 | 1.96% | yes |
| 2018-03 | 2.92% (25 of 855) | 0.91% (81 of 8,935) | 0.0202 | 6.46% | yes |
| 2018-04 | 0.92% (16 of 1,737) | 0.59% (78 of 13,326) | 0.0034 | 9.63% | yes |
| 2018-05 | 1.36% (26 of 1,909) | 0.85% (144 of 17,030) | 0.0052 | 12.31% | yes |
| 2018-06 | 2.12% (48 of 2,261) | 1.14% (199 of 17,487) | 0.0098 | 12.64% | yes |
| 2018-07 | 1.81% (36 of 1,991) | 1.25% (225 of 18,028) | 0.0056 | 13.03% | yes |
| 2018-08 | 1.80% (44 of 2,442) | 1.00% (191 of 19,168) | 0.0081 | 13.85% | yes |
| 2018-09 | 2.44% (55 of 2,253) | 0.90% (151 of 16,749) | 0.0154 | 12.10% | yes |
| 2018-10 | 2.01% (48 of 2,387) | 0.98% (164 of 16,806) | 0.0104 | 12.15% | yes |
| 2018-11 | 2.22% (27 of 1,218) | 1.15% (91 of 7,885) | 0.0106 | 5.70% | yes |

**X2, H2 by spend area** (standardised difference 0.0103, [0.0070, 0.0137])

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| Additives | 1.82% (22 of 1,207) | 1.03% (93 of 8,999) | 0.0079 | 6.50% | yes |
| CAPEX & SOCS | 2.16% (6 of 278) | 1.46% (64 of 4,394) | 0.0070 | 3.18% | yes |
| Chemicals & Intermediates | n/a (0 of 0) | 0.00% (0 of 1) | n/a | 0.00% | no |
| Commodity Resins | 1.25% (1 of 80) | 2.21% (5 of 226) | -0.0096 | 0.16% | yes |
| Energy | n/a (0 of 0) | 0.00% (0 of 3) | n/a | 0.00% | no |
| Enterprise Services | 0.00% (0 of 20) | 2.00% (7 of 350) | -0.0200 | 0.25% | yes |
| Latex & Monomers | 1.88% (9 of 479) | 2.09% (44 of 2,105) | -0.0021 | 1.52% | yes |
| Logistics | 35.96% (41 of 114) | 10.49% (205 of 1,955) | 0.2548 | 1.41% | yes |
| Marketing | 12.04% (13 of 108) | 3.06% (34 of 1,112) | 0.0898 | 0.80% | yes |
| Others | 5.81% (5 of 86) | 1.54% (26 of 1,690) | 0.0428 | 1.22% | yes |
| Packaging | 1.10% (76 of 6,901) | 0.46% (263 of 56,660) | 0.0064 | 40.95% | yes |
| Pigments & Colorants | 2.71% (8 of 295) | 1.29% (18 of 1,399) | 0.0143 | 1.01% | yes |

| stratum | failing-group O rate | comparison O rate | difference | comparison weight | kept |
|---|---|---|---|---|---|
| Real Estate | 0.00% (0 of 5) | 2.05% (3 of 146) | -0.0205 | 0.11% | yes |
| Sales | 1.02% (33 of 3,244) | 0.78% (340 of 43,647) | 0.0024 | 31.54% | yes |
| Solvents | 3.24% (7 of 216) | 1.55% (21 of 1,357) | 0.0169 | 0.98% | yes |
| Specialty Resins | 3.51% (8 of 228) | 1.03% (11 of 1,067) | 0.0248 | 0.77% | yes |
| Spend Area Unidentified | 0.00% (0 of 3) | 0.00% (0 of 15) | 0.0000 | 0.01% | yes |
| Titanium Dioxides | 7.56% (9 of 119) | 2.58% (8 of 310) | 0.0498 | 0.22% | yes |
| Trading & End Products | 2.13% (71 of 3,339) | 1.22% (143 of 11,765) | 0.0091 | 8.50% | yes |
| Workforce Services | 17.74% (11 of 62) | 16.89% (37 of 219) | 0.0085 | 0.16% | yes |
| missing | 1.88% (7 of 373) | 1.04% (10 of 957) | 0.0083 | 0.69% | yes |

### X3, W = 30 days, by item category

| category | gate holds | without a corrective | clustered 95% | share of held value | clustered 95% |
|---|---|---|---|---|---|
| 2-way | 4 | 100.00% | [100.00%, 100.00%] | 100.00% | [100.00%, 100.00%] |
| 3-way, invoice after receipt | 1,592 | 96.11% | [94.22%, 97.86%] | 81.29% | [69.13%, 92.04%] |
| 3-way, invoice before receipt | 16,152 | 98.90% | [98.69%, 99.10%] | 97.64% | [96.49%, 98.58%] |

### X3, W = 60 days, by item category

| category | gate holds | without a corrective | clustered 95% | share of held value | clustered 95% |
|---|---|---|---|---|---|
| 2-way | 4 | 100.00% | [100.00%, 100.00%] | 100.00% | [100.00%, 100.00%] |
| 3-way, invoice after receipt | 1,592 | 93.15% | [90.64%, 95.59%] | 68.86% | [58.57%, 80.18%] |
| 3-way, invoice before receipt | 16,152 | 98.40% | [98.13%, 98.65%] | 96.59% | [95.31%, 97.68%] |

### X3, W = 30 days, by line-value quartile

| quartile | gate holds | without a corrective | clustered 95% | share of held value | clustered 95% |
|---|---|---|---|---|---|
| Q1 | 2,334 | 99.44% | [99.03%, 99.78%] | 99.32% | [98.69%, 99.81%] |
| Q2 | 2,724 | 99.45% | [99.14%, 99.73%] | 99.45% | [99.13%, 99.74%] |
| Q3 | 4,522 | 99.12% | [98.77%, 99.43%] | 99.08% | [98.72%, 99.41%] |
| Q4 | 8,168 | 97.89% | [97.38%, 98.37%] | 91.76% | [86.19%, 95.93%] |

### X3, W = 60 days, by line-value quartile

| quartile | gate holds | without a corrective | clustered 95% | share of held value | clustered 95% |
|---|---|---|---|---|---|
| Q1 | 2,334 | 99.14% | [98.65%, 99.56%] | 98.94% | [98.22%, 99.55%] |
| Q2 | 2,724 | 99.08% | [98.62%, 99.49%] | 99.03% | [98.52%, 99.48%] |
| Q3 | 4,522 | 98.81% | [98.41%, 99.16%] | 98.73% | [98.30%, 99.12%] |
| Q4 | 8,168 | 96.71% | [96.02%, 97.35%] | 86.63% | [81.19%, 91.60%] |

Sources: `results/exploratory/d4-sensitivity.json`, `results/exploratory/adjusted-h1-h2.json`, `results/exploratory/hold-friction.json` (exploratory).